

Behavioural Responses to Tax Systems: Evidence from Costa Rica's Corporate Tax Reform

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Abstract

We study firms' responses to Costa Rica's 2019 CIT reform, which replaced a notched schedule with a kinked one, reducing distortions in the system at the cost of simplicity. Using 2016–2023 administrative returns in a difference-in-differences design that exploits rich variation in marginal and average rates, we find: (i) signs of tax manipulation, such as bunching and profit-margin discontinuities at the first (now eliminated) revenue threshold, declined promptly after the reform; (ii) responses were asymmetric, with reactions to tax increases weaker than those to tax cuts, likely due to the removal of growth barriers; (iii) firms behaved like profit maximisers: their profits and other intensive-margin variables responded to marginal but not average tax rates, in contrast with evidence that individuals, when confronted with relatively complex non-linear schedules, conflate the two rates and deviate from optimising behaviour. Overall, clear communication of complex reforms can improve efficiency without raising compliance errors.

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1. Introduction

When designing tax policies, it is important to consider key desirable characteristics such as simplicity, efficiency, redistribution and revenue collection, as these objectives can sometimes be in tension (OECD 2010). In corporate income taxation, linear flat tax structures are recognised for their simplicity and efficiency, leading to widespread adoption. Yet, concerns over redistribution have also motivated some countries to implement progressive tax designs (OECD 2022). Under a progressive corporate income tax (CIT), policymakers must navigate the trade-off between simplicity and efficiency inherent in non-linear schedules.

This paper examines the major 2019 CIT reform in Costa Rica, a country characterised by a progressive tax structure, to analyse how firms responded to a policy change that reduced distortions at the expense of reduced simplicity. By eliminating clear inefficiencies, the reform also introduced intricacies and multiple adjustments that may have made optimal compliance more challenging for affected firms to quickly and fully understand. To evaluate its effects, we exploit rich variations in marginal and average tax rates as identifying changes.

Our analysis reveals two main findings. First, we identify asymmetric responses to tax changes: firms reacted more strongly to tax cuts than to tax increases. This may reflect the elimination of growth barriers on declared revenues for small firms, or the role of evasion costs and perceived audit risks that make downward adjustments to reported profits more costly. Second, we find no evidence of behavioural errors. Firms adjusted their tax reports rapidly and in line with predictions from after-tax profit maximisation models, contrasting with prior evidence for individuals, who tend to deviate from optimal responses under relatively complex income tax systems (Rees-Jones and Taubinsky 2020; De Bartolome 1995).

In contexts characterised by limited enforcement capacity, such as Costa Rica, tax simplicity has been often prioritised over efficiency to reduce administrative burdens for both tax authorities and taxpayers. Moreover, simplicity has shown to help limit tax manipulation, with empirical evidence indicating that simpler corporate tax systems can reduce opportunities for evasion (Best et al. 2015). Simplicity is also valued for its clarity, making tax schedules easier to interpret and potentially minimising the risk

of taxpayers deviating from optimal compliance—a challenge observed in developing economies (Almunia et al. 2024; Tourek 2022). These advantages of simplicity partly explain the adoption of notched structures in tax and pricing systems despite their efficiency costs (Kleven and Waseem 2013), as was the case with the pre-reform CIT analysed in this study. Notches, which introduce discontinuities in average tax rates rather than marginal tax rates (as in systems with kinks), are generally considered more salient and easier for taxpayers to navigate (Slemrod 2013). In notched systems, taxpayers only need to identify the single relevant tax rate applicable to their tax base to calculate their liability, rather than managing multiple rates as required in kinked systems. Nonetheless, by creating discontinuities in average tax rates, they may also trigger stronger behavioural responses compared to kinks (Kleven 2016)

Before the implementation of Costa Rica’s CIT reform in October 2019, at the beginning of the 2020 fiscal year, the corporate tax system applied a simple and progressive structure of increasing average tax rates as a function of firms’ reported revenue levels, levied on reported profits. This schedule had average tax rates that jumped abruptly from 10% to 20% at the first revenue threshold and from 20% to 30% at the second. These sharp discontinuities created strong incentives for firms to report revenues just below the revenue thresholds, while those exceeding the thresholds inflated costs to reduce their taxable income (Bachas and Soto 2021). To partially address these inefficiencies, the 2019 reform introduced a profit-dependent, multiple kinked schedule that smoothed the first jump in the average tax rate for small and medium firms. This meant that the first revenue-dependent threshold was removed, a redesign aimed to lower manipulation incentives for implicated firms. Through this, the reform introduced differentiated adjustments to marginal and average tax rates, creating gaps between these two rates. In contrast, the reform brought no change for large firms, both their marginal and average tax rates remained fixed at 30%, with the purpose of maintaining higher rates for these firms, leaving the second revenue threshold in place.

We utilise 2016-2023 corporate tax return panel data to track corporate tax reports before and after the CIT reform. Our descriptive analysis reveals reduced revenue and costs manipulation. We find that the previous bunching around the first revenue threshold disappears, resulting in a smooth revenue distribution consistent with a decline in tax reports’ manipulation behaviour that was documented before the reform (Bachas and Soto 2021). Additionally, firms past the first threshold no longer show a profit margin

discontinuity pattern previously indicative of costs inflation. These patterns suggest that, by removing this source of distortion, evasion responses for firms around the first (now eliminated) threshold markedly decreased following the corporate tax reform. However, for firms operating around the second (still present) threshold, we continue to observe persistent bunching and an even more pronounced discontinuity in profit margin declarations.

Since the reform changed firms' tax rates differently based on their position in the reported revenue and profits distributions, we exploit this variation to implement a difference-in-differences approach, allowing us to infer causal behavioral responses. We estimate the impact of marginal and average tax rate variation on firms' intensive margin outcomes, focusing primarily on reported profits, while also examining reported revenues and costs. To mitigate endogeneity concerns, we use the mechanical tax rate changes to define the treatments and to construct an instrument for the actual net-of-tax rates, defined as the tax rate a firm would face if it continued to report its pre-reform tax information.

Our causal analysis reveals that the elasticity of reported profits with respect to the marginal net-of-tax rate is on average large, at 4.7%, consistent with previous estimates for Costa Rica and other developing economies (Bachas and Soto 2021; Tourek 2022; Alejos 2018). The unique structure of the reform, which introduced both tax increases and reductions for similar size firms, allows us to estimate responses to differently signed tax changes within the same policy event. Our results show that firms experiencing a reduction in their marginal tax rate reported higher profits relative to the control group, while firms facing an increase reported lower profits, consistent with standard after-tax profit maximisation. We also document that these responses were asymmetric: reactions to tax cuts were much larger than those to tax increases. This asymmetry may reflect the removal of growth barriers for small firms, or the role of evasion costs and perceived audit risks that make reducing reported profits more costly. Regarding the anatomy of these responses, we find that firms facing lower marginal tax rates primarily increased reported profits through higher revenue declarations, while those experiencing higher rates adjusted mainly through reported costs.

Furthermore, we find that firms' reported profits responded rapidly to marginal tax rates as expected theoretically, and they do not seem to be sensitive to average tax

rates changes. This result contrasts with prior research suggesting that individuals, when confronted with relatively complex, non-linear individual income tax schedules, deviate from optimising behaviour (Rees-Jones and Taubinsky 2020; De Bartolome 1995). This finding implies that well-communicated tax reforms, such as the Costa Rican CIT reform—despite its relative complexity—can help tax authorities implement less distortive systems, while minimising errors.

This study makes several contributions to the literature. First, it adds to the expanding body of research on firm income taxation in low- and middle-income countries, which has highlighted the significant responsiveness of firms to corporate taxes (Lobel et al. 2024; Bachas and Soto 2021; Alejos 2018; Waseem 2018). This responsiveness has been argued to be largely driven by the common challenge of low enforcement capacity in these settings, which creates a larger room for evasion. Unlike most existing studies, which primarily focus on local estimates of tax elasticities obtained using bunching techniques, this paper provides one of the first non-local corporate elasticity estimates in a developing country context by leveraging a reform and a difference-in-differences approach.

Secondly, this paper contributes to the literature on asymmetric responses to taxes by documenting stronger reactions to tax reductions than to tax increases. Asymmetric responses to differently signed tax changes have been documented for VAT, payroll taxes, individual income taxes, and wealth taxes (Benzarti et al. 2020; Saez et al. 2019, 2021; Hussain and Malik 2016; Wu and Yang 2024). Evidence also exists on the effects of CIT on output and consumption from a macroeconomic perspective (Hussain and Malik 2016), though this study provides insights using a microeconomic approach. Moreover, the unique design of the reform allows this to be the first study, to the best of our knowledge, to estimate simultaneously asymmetric tax responses under a single reform event, and not two or more as in other studies, ensuring that observed differences are not driven by circumstantial factors.

Finally, this paper contributes to the literature on optimising behaviour in the context of progressive non-linear tax schedules. While prior research highlights suboptimal responses among consumers and individual taxpayers facing relatively complex kinked structures (Ho 2023; Rees-Jones and Taubinsky 2020; Shaffer 2020; Ito 2014; De Bartolome 1995), as well as firms making costly errors even with simpler tax systems in

developing countries (Almunia et al. 2024; Tourek 2022), this study finds no evidence of suboptimal behaviour among firms operating under an arguably harder to understand CIT schedule, and despite multiple changes in the tax design. Instead, we find that firms in Costa Rica appear to act as profit-maximising agents, with their profits declarations responding quickly and rationally to changes in marginal tax rates without their intensive margin outcomes being significantly influenced by average tax rates. This is the first study to examine this for firms, contrasting with findings for other economic agents who have been shown to oversimplify or *iron* kinked tax schedules due to the challenges or costs of responding to more complex incentives, reacting to average tax rates where models of optimal behaviour predict responses to marginal taxes (Rees-Jones and Taubinsky 2020; Liebman and Zeckhauser 2004; De Bartolome 1995).

This document is organised as follows. Section 2 introduces and summarises the data used for our analysis. Section 3 describes Costa Rica’s corporate tax schedule both before and after the tax reform and presents key descriptive patterns. Section 4 details the empirical strategy employed in this study. Section 5 presents our empirical findings and checks. Finally, Section 6 concludes, discussing policy implications and next steps.

2. Data

We utilise the 2016-2023 Registry of Economic Variables (Revec), a dataset compiled by the Central Bank of Costa Rica. This dataset contains the complete set of administrative corporate tax returns submitted to the Ministry of Finance. All registered and formal firms in Costa Rica are required to file an electronic annual tax declaration (D-101)¹, which includes typical balance sheet information and tax reports such as reported revenues, costs, and profits. This data is used to calculate corporate taxes based on the prevailing tax structure, as described in Section 3. In addition to tax returns, the Central Bank of Costa Rica enhances the dataset with firm-level information, such as employment, industry classification, and international trade transactions. These supplementary data are sourced from administrative records, including social security,

¹Firms under the simplified taxation regime follow a different schedule and file a distinct tax declaration for income tax purposes, and thus are not included in our analysis. To be part of this regime, firms must meet specific revenue and industry criteria. Eligible firms may apply for entry with the tax authority.

customs, and the national firm registry. Furthermore, we calculate additional variables like the profit margin (defined as reported profits over reported revenues) using this dataset.

For this study, we focus on firms that have records on reported revenue and costs data. We work with firms that with positive revenue below 350,000,000 CRC, normalised relative to the second tax reported revenue threshold established in 2014 ². We exclude firms that never report revenues exceeding 3,000,000 CRC during the analysis period to eliminate those with minimal activity. Finally, we clean the dataset by trimming the profit margin variable within the 1st and 99th percentiles of 1 million CRC-year bins.

3. Institutional Setting

Costa Rica enacted a significant corporate tax reform with the approval of the Law on the Strengthening of Public Finances by Congress on December 3, 2018. ³ The CIT reform was implemented on October 1, 2019, marking the start of the 2020 fiscal year⁴, as an effort to decrease clear tax manipulation incentives present in the previous tax design.

Prior to the 2020 fiscal year, Costa Rica imposed an increasing average tax rate (*ATR*) on reported profits, defined as reported revenue minus deductible costs. While the statutory rate was 30%, smaller size firms benefited from two preferential lower tax rates based on their reported revenues. As illustrated in Figure 1, both the *ATR* and marginal tax rate (*MTR*) exhibited significant jumps around two reported revenue thresholds under a notched tax schedule. Therefore, the higher tax rate when surpassing a reported revenue threshold applied to all reported profits, not just the portion exceeding the threshold. At the first reported revenue threshold (T_1 , equal to annual reported rev-

²Since the thresholds are updated annually, we re-center the data by normalising each year's observations to the second reported revenue threshold of 2014, ensuring that all thresholds remain constant over time. This transformation allows us to consistently compare bunching behaviour across different years. We utilise this normalised data for data cleaning and to produce descriptive graphs.

³This comprehensive reform package also replaced the sales tax with a value-added tax, established a fiscal rule to limit public spending, and restructured the salary framework for public employees.

⁴Costa Rica also redefined its fiscal year. Before the reform, the fiscal year spanned from October to September. With the reform, the fiscal year changed to align with the calendar year, running from January to December, and starting from the 2021 fiscal year onwards. The 2020 fiscal year was a transitional year, extended for 15 months from October 2018 to December 2019.

enues of 49,969,000 CRC or approximately 94,000 USD in 2014), the *ATR* jumped from the 10% that was applied to those considered as small firms, to a 20% to the medium-sized firms. At the second threshold (T_2 , equal to 100,513,000 CRC or approximately 189,000 USD in 2014), the *ATR* increased further from 20% to the statutory 30% rate for large firms. Under this design, the determinant of the tax rate (reported revenue) differed from that of the tax base (reported profits), and the *ATR* was inherently equal to the *MTR*.

[Figure 1 here]

The design of notched tax systems, as analysed by Kleven and Waseem (2013), Londoño-Vélez and Avila-Mahecha (2024), and other related literature, is known to be highly distortionary. Firms subject to these systems face strong incentives to relocate from just above the threshold to just below it in order to qualify for a lower tax rate. In the case of the now-extinct CIT system, this behavior led to a clear excess masses of firms bunching just above the thresholds, as well as indicatives of missing masses just after. Figure 2A illustrates this, showing the distribution of firm density for three years preceding the reform (2016, 2017 and 2018). Additionally, tax reports from firms reveal patterns consistent with tax evasion, with reported profit margins displaying notable discontinuities, indicative of substantial evasion responses to the notched design (see Figure 2B). Specifically, the three step-down pattern in the distribution suggests that firms slightly above the thresholds under-reported revenue to achieve a lower *ATR*, while firms already above the thresholds inflated costs to reduce their profit margin and hence their tax liability.

[Figure 2 here]

3.1. CIT After the Reform

After the 2019 CIT reform, large taxpayers—those with reported revenue exceeding the second reported revenue threshold (T_2)—continue to face a statutory *ATR* of 30% on reported profits. For other firms, the tax schedule underwent significant changes in three major ways. First, the preferential tax rates are now directly based on reported profits rather than being exclusively determined by reported revenue. Second, the previously first sharp jump in the *ATR* has been smoothed out. Under a new schedule with kinks, higher *MTRs* are applied only to the excess profits above the corresponding thresholds.

Third, the schedule has been redefined with new profit thresholds and new *MTRs*. These changes are summarised in Figure 3. Panel A shows that the *MTR* for small and medium-sized firms ranges from 5% to 20%, determined by three new profit-dependent thresholds: t_1 (equal to 4,741,000 CRC), t_2 (7,112,000 CRC), and t_3 (9,482,000 CRC)⁵. Panel B shows that for large firms, the *MTR* (and the *ATR*) remained at 30% after the reform. Essentially, the reform eliminated the first reported revenue-dependent notch but retained the second threshold, which keeps marking the start of the still statutory 30% *ATR* for large firms. Table 1 summarises the main features of the CIT before and after the reform.

[Table 1 here]

[Figure 3 here]

Descriptive analysis reveals large and rapid changes in reporting behavior in the years following the reform, 2021, 2022 and 2023. First, Figure 4A shows the disappearance of excess and missing mass around the extinct first notch, aligning with the removal of incentives to bunch at that point. Conversely, we second observe that bunching remains evident at the second threshold, which still separates small and medium firms from large ones. The incentives to bunch at this upper reported revenue threshold persist and have even increased, as the *MTR* and *ATR* before this threshold decreased for many firms. Specifically, before the reform, previously medium firms crossing the second threshold experienced a jump of 10pp in their *MTR*, from 20% to 30%. After the reform, crossing this threshold can now result in a jump of up to 25pp in *MTR* for lower profitability firms. Hence, as we will discuss more below, there is also a stronger incentive to decrease declared profitability when passing this point.

Third, we observe what appears to be a decrease in firm density for the bins below the first notch and an increase for the bins above it, suggesting that the first notch may have disincentivised firms from growing—or at least from reporting higher revenues to the tax administration. This trend becomes clearer when examining the density changes around the first threshold, as depicted in Figure 5A. Here, we observe that with the reform, firm density sharply decreases before and increases after this removed threshold. However, we do not observe a similar pattern around the second threshold that remained in place, as seen in Figure 5B. Instead, we see a general increase in firm

⁵Values expressed in 2014 CRC.

density in that vicinity. It is important to consider that the decrease in density before the first threshold and the increase after could also be at least partially explained by the pandemic, during which the economic crisis disproportionately affected small firms, leading to their disappearance and the survival of more medium and larger firms.

[Figure 4 here]

[Figure 5 here]

The different incentives also led to changes in the profit margin behavior pattern, as seen in Figure 4B. The previously observed discontinuity in profit margins at the first threshold vanishes, leading to noticeably higher average profit margins for medium firms. Meanwhile, the profit margin discontinuity around the still-present second threshold becomes more pronounced, consistent with the higher incentives to lower profit margins when crossing the threshold. Figure 4B shows that while smaller firms do not seem to have increased their profit margins, medium firms appear to have done so significantly, and large firms also, although to a lesser extent. The role of costs declarations in this pattern becomes clear in Figure 4C. After the reform, the jump in costs declarations smooths out at the first threshold and becomes much sharper at the second one. All these descriptive patterns strongly suggest pronounced and prompt behavioral responses to the CIT reform. Additionally, average employment depicted in Figure 4D reflects that before and after the reform, employment's first and second moments seem to grow steadily with reported revenue, with no particular changes occurring near either of the two thresholds. This suggests that the discontinuities observed in density, reported profit margins, and reported costs around the thresholds are importantly explained by evasion responses rather than real adjustments.

3.2. Variation in Tax Rates

By altering the tax system design, the reform also introduced heterogeneous tax rate changes across firms. Due to the reduction in discontinuities in the *ATR* from the removal of the notch and the introduction of new profit thresholds with new tax rates, firms across the profit distribution could have experienced a range of increases or decreases in their *MTR* and *ATR*. For instance, a firm could face a change in their *MTR* from 10% or 20%, to either 5%, 10%, 15%, or 20%, for the same tax report. Specifically,

and as shown in Figure 6, small firms with reported revenue up to the first reported revenue threshold (T_1) saw a decrease, no change, or an increase in their *MTR*. For medium firms with reported revenue between T_1 and T_2 , the *MTR* either decreased or remained unchanged. Lastly, large firms did not experience a change in their *MTR*. Analogously, the *ATR* could have change substantially for small and medium firms with the same tax report. The figure also shows the split we do for firms into seven categories, depending on how their *MTR* and *ATR* was affected by the reform, by firm size and reported profits.

[Figure 6 here]

4. Empirical Strategy

This section outlines how we leverage tax rate variations induced by the reform to investigate behavioural responses to corporate taxation. We employ difference-in-differences (DiD) research designs for our estimations, comparing the behavior of treated firms that experienced a change in their *MTR* due to the reform, and different changes in their *MTR* and *ATR* for later estimations, to those that did not, both before and after the reform.

As we focus on evaluating intensive margin responses to changes in tax rates, and to minimise potential biases arising from extensive margin responses (firm entry or exit), our primary specifications will be estimated using a balanced panel of firms. In this sample we also only include firms with complete information on employment variables that serve as controls in our analysis. Our main sample comprises 90,376 firm-year observations over the eight-year period, representing 11,297 unique firms. Table 2 provides summary statistics for this sample, disaggregated by firm size.

[Table 2 here]

4.1. Expected Responses to Tax Changes

We delineate which outcomes are expected to be affected by marginal and average tax rate changes. Previous theoretical and empirical research has broadly established that

intensive margin decisions, such as the level of production or the amount of profit declared to tax authorities, are driven by *MTR* faced by firms. In contrast, extensive margin or discrete decisions, such as entry-exit choices or investment location, are influenced by the *ATR* (Devereux and Griffith 2003).

In this paper, we analyse the effects of the tax changes intensive margin outcomes — focusing on reported profits, but also examining revenues, and costs— which are expected to respond to adjustments in the *MTR*. Most of our findings relate to how the tax base, reported profits, reacts to changes in the *MTR*. According to predictions from standard after-tax profit maximisation models, a reduction in the *MTR* leads to higher reported profits (Devereux et al. 2014; Best et al. 2015). This relationship can be explained by both real and evasion-related responses. Firstly, a lower *MTR* increases the marginal benefit of selling an additional unit, encouraging firms to expand real production and generate more revenue. Secondly, a lower *MTR* diminishes the incentive for tax evasion. Firms are less likely to under-report revenues or inflate costs when the potential benefits from evasion decrease. In contrast, when the *MTR* increases, firms face a higher tax burden, which not only decreases incentives for real production but also increases the motivation to underreport profits or inflate costs, resulting in lower expected reported profits overall.

4.2. Treatment Definition

We aim to identify the effects of tax rate changes induced by the reform on firms' behaviour. Directly using the actual changes in tax rates to define binary treatments, or the actual net-of-tax rates for elasticity estimates, is clearly problematic, as these are endogenous to the reform. Depending on the prevalent schedule, they are a function of declared revenues before the reform, and of reported revenue and reported profits after the reform, which are independent variables in our regressions. To address this issue, we follow the established literature and use the mechanical tax rates instead (Weber 2014).

Specifically, using the mechanical tax rate change allows us to use pre-reform information only to define treatment groups and to construct a continuous instrument for the actual net-of-tax rates faced by firms, as ways to capture individual firm's exposure to the tax changes introduced by the CIT reform. The mechanical tax change is defined

as the percentage point variation in the tax rate (*MTR* or *ATR*) that a firm would face after the reform, assuming it continued to report the same tax information as before the reform—specifically, the same reported revenue and profits. Through this study we define treated firms as those that experienced changes in their mechanical tax rates, while untreated firms are those that saw no change.

This method enables us to address endogeneity concerns and better isolate the causal impact of tax rate changes. To mitigate concerns about mean reversion, anticipation of the reform, idiosyncratic shocks, and firms self-selection to the treatment or control groups, we construct the mechanical tax rate changes using the full set of firms' 2016–2019 tax reports, rather than relying solely on the single-year 2019 pre-reform tax declaration. For example, a firm reporting an average of CRC 45 million in revenue and CRC 4 million in profits during the 2016–2019 pre-reform period would experience a change in its *MTR* from 10% to 5%, based on the tax schedules before and after the reform, as described in Section 3.

Panel A of Figure 7 illustrates the average, minimum and maximum of mechanical *MTR* changes in percentage points, calculated based on firms' average reported revenue and profits from 2016–2019, by average reported revenue bins. The figure indicates that a significant proportion of small and medium-sized firms experienced a decrease in their *MTR* under this definition, with an average reduction of 7pp, a minimum decrease of 5pp, and a maximum of 15pp, representing 47% of our balanced panel sample. Conversely, a smaller subset of small firms faced an increase in their *MTR*, with an average tax rise of 8pp, a minimum increase of 5pp, and a maximum of 10pp, accounting for 6% of the sample. The remainder of the sample consists of firms that according to our mechanical tax rate definition did not experience a change in their mechanical *MTR*, a group comprising small, medium, and large firms. Panel B presents the equivalent distribution for mechanical *ATR* changes. By construction, large firms were not treated under either classification.

[Figure 7 here]

As a first approach, we consider a binary treatment definition, D_i , where our treatment variable takes the value of one, $D_i = 1$, if the calculated mechanical marginal tax rate change of a firm is different from zero, $\Delta MTR \neq 0$, and zero otherwise. We use this

to compare firms that experienced a change in their *MTR* relative to the control group, consisting of firms with no mechanical tax rate change.

As a second strategy, we employ a continuous instrument for the net-of-tax rate to capture the variation in the degree of exposure to the CIT reforms. For observations before the reform, the value of this variable is equal to one minus the marginal tax rate ($1 - MTR$) under the previous tax schedule based on the average reported information from 2016-2019. From 2020 onwards, it takes the value of one minus the calculated new *MTR* after the reform, according again to the average tax reports from 2016-2019. We use this instrument to study the sensitivity of reported profits to the net-of-tax rates changes, and we also use it to explore its effects on reported revenue and costs to disentangle the profit responses.

Finally, we also integrate the *ATR* into our analysis. We perform a binary DiD to investigate how firms' reported profits responded to different combinations of changes in *MTR* and *ATR* for our six treatment groups shown in Figure 6. We also compute elasticity estimates for the different groups.

4.3. Estimation

Reported profits (π_{it}), the tax base and our main outcome of interest, are defined as the difference between reported revenue (y_{it}) and costs (c_{it}). Since firms report either positive or zero profits, we use Poisson quasi-maximum likelihood models throughout our analysis to account for the presence of zeros, as recommended by Silva and Tenreiro (2006) and Chen and Roth (2024). While most of our study focuses on the effects on reported profits, we also investigate the impacts on reported revenue and costs to better understand the anatomy of the observed responses.

We begin by employing a standard binary DiD design. Utilising a binary treatment enables us to causally identify average responses under a "standard" parallel trends assumption, without requiring the stronger version of the assumption required for a continuous treatment (Callaway et al. 2024). Our initial model is specified as follows:

$$(1) \quad \log(Z_{it}) = \alpha_i + \gamma_t + D_i \times \text{Post}\beta + X_i' \phi + v_{it}$$

Here, Z_{it} (reported profits, revenue or costs) denotes the outcome of interest for firm i in year t . α_i and γ_t represent individual and time fixed effects, respectively. The vector X'_i includes year fixed effects interacted by pre-reform industry, average wage bill bins (a proxy for firm size), and average wage bill to reported revenue ratio bins (a proxy for firm technology). D_i is a dummy variable indicating whether firm i was treated with the introduction of the reform, and Post_t is a dummy equal to 1 for the years t following the reform, according to our treatment definition described in the subsection above. The year fixed effects help us control for macro-level shocks that could have affected firms' real and reporting behaviour, such as the COVID-19 pandemic. Moreover, the interaction between year and pre-reform industry fixed effects further accounts for the fact that different industries may have been affected differently by such shocks. We assume that firms within the same industry were similarly impacted. Standard errors are clustered at the firm level. Our first analyses captures responses to MTR , as the reflected in the notation below, but we also preform similar specifications to incorporate investigations to the responses to ATR .

Our second set of specifications considers the fact that the reform may have lead to differential responses by whether tax rates decreased or increased. Hence, we perform separate binary DiD estimations for the group treated with a decrease in their MTR , $D_{\downarrow MTR,i}$ and the group experiencing an increase $D_{\uparrow MTR,i}$, using the same control group of not treated firms in both estimations:

$$(2) \quad \begin{aligned} \log(Z_{it}) &= \alpha_i + \gamma_t + D_{\downarrow MTR,i} \times \text{Post}_t \beta + X'_i \phi + v_{it}, \\ \log(Z_{it}) &= \alpha_i + \gamma_t + D_{\uparrow MTR,i} \times \text{Post}_t \beta + X'_i \phi + v_{it} \end{aligned}$$

Additionally, we analyse firms' behaviour over time around the introduction of the reform using an event-study specification of the DiD approach and look at the dynamics of the effects:

$$(3) \quad \begin{aligned} \log(Z_{it}) &= \alpha_i + \gamma_t + \sum_{\kappa=-4}^{-2} \theta_{\kappa} D_{\downarrow MTR,i} + \sum_{\kappa=0}^3 \delta_{\kappa} D_{\downarrow MTR,i} + X'_i \phi + v_{it}, \\ \log(Z_{it}) &= \alpha_i + \gamma_t + \sum_{\kappa=-4}^{-2} \theta_{\kappa} D_{\uparrow MTR,i} + \sum_{\kappa=0}^3 \delta_{\kappa} D_{\uparrow MTR,i} + X'_i \phi + v_{it} \end{aligned}$$

Here the coefficients $\{\theta_k\}_{k=-4}^{-2}$ are the event-study DiD pre-trends and $\{\delta_k\}_{k=0}^3$ are the treatment effects relative to 2019 (corresponding to the omitted period -1).

Finally, we assess the effects of varying treatment intensities on firms' behavior and derive elasticity estimates using the instrument for the continuous net-of-tax rate. To this, we use the following specification:

$$(4) \quad \log(Z_{it}) = \alpha_i + \gamma_t + \varepsilon_{1-MTR} \cdot (1 - MTR_{it}) + X'_i \phi + v_{it}$$

5. Results

5.1. Effects on Reported Profits (Tax Base)

5.1.1. DiD Estimations

The coefficients in line 1 of Panel A in Table 3 present the estimated average effect of being treated by a change in MTR on reported profits, relative to the control group, as derived from the binary DiD analysis. In this part of the analysis, our treatment group consists of small and medium firms that experienced a change in their MTR , while our control group comprises small, medium, and large firms that saw no change in their MTR . Given that we use an instrument for treatment and that the coefficients from the models are not directly interpretable, we compute the implied intention-to-treat estimates (ITT) using the formula $ITT = \exp(\beta) - 1$ (Chen and Roth 2024). The adjusted estimates, which we focus on for interpretation, are reported in line 3.

[Table 3 here]

For comparability, Columns 1–3 of Panel A present results for the unbalanced panel. This dataset includes firms with observations for at least the years 2019 and 2020, i.e., firms that were affected by the reform and for which we construct the treatment, but

some could have entered or exited the sample through our analysis period ⁶. Columns 4–6 display results for our preferred balanced-panel sample, but without controls. Finally, Columns 7–9 report results for our preferred specification, both for this table and for other upcoming results, which uses the balanced-panel sample and includes controls.

Column 7 of Panel A shows that firms affected by a change in *MTR* (whether an increase or decrease) experienced an average rise of 36.6% in reported profits post-reform compared to pre-reform levels, indicating a general increase in reported profits among treated firms. Columns 8 and 9 further distinguish responses based on the direction of the *MTR* change. Firms facing a lower *MTR* reported an average profit increase of 61.2% (corresponding to an average tax rate reduction of 7pp), while those subject to a higher *MTR* experienced a 13.2% decline (associated with an average tax rate increase of 8pp). These results align with standard after-tax profit maximisation models and the expected effects of tax changes, though the response is notably stronger for firms treated with a reduced *MTR*. Results from the balanced panel without controls (Columns 4 to 6) exhibit similar patterns but with slightly larger coefficients. This suggests that controlling for firm size and technology attenuates the estimated treatment effects.

Looking at Columns 1 to 3, the results using the unbalanced panel sample are qualitatively similar to those of our preferred specifications but exhibit even larger magnitudes than those in Columns 4 to 6. This difference arises because the estimates also now include firms that either entered or exited the sample during the analysis period. For firms that, based on their reported information and our treatment definition, would have received a decrease in their *MTR*, the larger estimated positive effects may be driven by a proportion of newly entering firms that are relatively more profitable. These firms might have chosen to register under the reduced rates, and given their actual or potential levels of revenue and profits, either as new businesses or previously informal firms that now find formal registration viable. Moreover, those exiting, giving COVID or other idiosyncratic shocks, could be those relatively less profitable. In the case of firms that, based on their reported information and treatment, would have received an increase in their *MTR*, the larger estimated negative effects could be driven by less profitable firms entering and more profitable firms exiting.

⁶See Table A1 for descriptive statistics of this alternative sample.

It is important to note that the treatment definition using a binary DiD considers only whether firms experienced a change in their marginal tax rates, without accounting for the magnitude of the change. As such, we cannot draw strong conclusions about firms' sensitivity to the size of tax changes. To address this limitation, we present elasticity estimates in Subsection 5.1.3. Nonetheless, these results suggest asymmetric responses to changes in marginal tax rates: firms appear to react more strongly to tax decreases than to tax increases.

5.1.2. Dynamic Effects: DiD Event-Study Estimations

Figure 8 illustrates the dynamics of the findings using the event-study specification of the DiD approach. Panel A shows that firms experiencing a decrease in *MTR* followed parallel trends with the control group prior to the policy change. Following the reform's enactment, there is a sharp and immediate increase in reported profits, which persists over time, although with a gradual deceleration. In contrast, Panel B shows that firms subject to an increased *MTR* also exhibited parallel trends in reported profits before the reform and experienced a decline in reported profits in the year following the reform (2020). However, the magnitude of this decline is smaller compared to the increase observed for firms with a reduced *MTR*. Furthermore, firms facing these tax change do not exhibit significant effects on reported profits in subsequent years, though a non-significant downward trend appears to persist. These results align with what was discussed for the binary DiD specifications.

[Figure 8 here]

Additionally, Figure A1 presents results for the unbalanced sample without controls. These results are qualitatively consistent with those from our preferred dynamic specifications but exhibit stronger effects. The biggest difference arises for firms experiencing an increase in *MTR*, where the decline in reported profits appears more persistent and intensifies over time. As previously discussed, the potential impact of firm entry and exit on our estimates applies to these specifications as well.

5.1.3. Elasticity Estimates

Columns 7–9 of Panel B in Table 3 present the reported profit elasticity estimates with respect to changes in the marginal net-of-tax rate, providing a more precise measure of firms' sensitivity to tax changes. Using the full sample, the results indicate that a 1% increase in the net-of-tax rate leads to an average increase of 3.9% in reported profits. Although this elasticity is large compared to estimates from high-income countries, where values are typically below 1 (Devereux et al. 2014), our findings are consistent with the hypothesis and empirical evidence that elasticities tend to be significantly higher in developing economies with weaker enforcement capacity (Tourek 2022; Alejos 2018; Patel et al. 2017). Moreover, our results align with previous local estimates from Costa Rica under the old tax schedule, derived using bunching techniques, which estimate the elasticities between 2.7% and 4.8% (Bachas and Soto 2021).

When focusing on the group that experienced a decrease in *MTR* relative to the control group, the estimated elasticity is 4.5%. In contrast, firms facing an increase in their *MTR* exhibit a lower elasticity estimate of 1.7%. This pattern aligns with findings in the literature on other types of taxes, which show that agents respond asymmetrically to tax increases and decreases (Benzarti et al. 2020; Saez et al. 2019, 2021; Hussain and Malik 2016; Wu and Yang 2024).

These results reinforce the asymmetry in responses depending on the direction of the tax change, as discussed in Subsection 5.1.1: reactions to tax cuts are larger than those to tax increases. We consider two potential explanations for this pattern. The first is that the reform may have eliminated growth barriers on declared revenues, particularly for small firms below T_1 , including those that experienced an increase in their *MTR*. This could have mitigated the expected downward adjustment in declared profits in response to higher tax rates. However, even for these firms, the adjustment in declared costs was still larger than the adjustment in declared revenues.

A second possible explanation is that asymmetries are linked to evasion costs. Firms may perceive that sharply reducing reported profits following an increase in the *MTR* raises the likelihood of an audit and the risk of penalties, which could lead to downward stickiness in profit reporting. While the first explanation appears more consistent with the observed patterns, we are currently working on further analysis to determine which mechanisms are driving the results.

The unique design of the reform allows this study, to our knowledge, to be the first to simultaneously estimate asymmetric tax responses within a single reform event, rather than across multiple tax changes at different points in time, as done in previous research. This ensures that observed asymmetric responses are not driven by time-specific factors.

5.1.4. Heterogeneity Analysis

Regarding the heterogeneity analysis, our current data availability allows us to perform it across firm size and industries. If additional data becomes available, we aim to extend this analysis to explore heterogeneity by geography and type of ownership. Currently, we perform this analysis using triple-difference specifications⁷.

We first examine heterogeneity by pre-reform firm size. To do this, we implement a triple-difference specification by adding an interaction term between the treatment effect and dummies indicating different firm sizes. Specifically, we divide the potentially treated small and medium firms into four groups based on pre-reform revenues. The cut-off points for these groups are $T1/2$, $T1$, $T1 + (T2 - T1)/2$, and $T2$, where $T1$ and $T2$ are the CIT revenue thresholds before the reform. This yields the following groups: *smaller small*, *larger small*, *smaller medium*, and *larger medium*. The base (excluded) category is the first one.

The treatment effect D reported in the first line of results in Table A2 reflects the effect for smaller small firms, while the below triple-difference coefficients indicate how that effect changes relative to this baseline across the other size groups. Looking at our preferred specifications — particularly column 8, which focuses on firms that saw a decrease in their *MTR* — the results show that the positive effect is equally large for both small firm categories. For medium firms, the effect remains positive, but its magnitude is smaller. In the case of *MTR* increases, shown in column 9 (note that the last two medium firm size categories are excluded here, as only small firms experienced *MTR* decreases; see Figure 7), the negative effect is largest for smaller small firms, while it is somewhat mitigated for the larger small firms. Overall, the evidence suggests that the smallest firms experienced the strongest responses to the reform.

⁷We plan to include further heterogeneity analysis using random forest methods in the next versions of the paper.

Secondly, we perform a heterogeneity analysis by the eleven industry categories. The base (excluded) category is education, human health & social work. Thus, the treatment effect D in the first line of results in Table A3 reflects the effect for the base industry, while the triple-difference coefficients show how that effect varies relative to this category across the other industries. Column 8, which shows the effect of the decrease in MTR , indicates that the base category saw a positive and significant effect. The industry categories of agriculture, forestry & fishing; construction; financial & insurance; real estate; and other services & NA appear to have a larger positive effect. In particular, agriculture, forestry & fishing and financial & insurance are the categories where the effect seems strongest. The other categories do not show significant differences compared to the baseline.

Column 9 shows the effects for firms that saw an increase in their MTR . Here, the effect is negative and significant for the baseline category education, human health & social work. Other categories do not display significant differences with the baseline, except for the professional, scientific, technical & administrative category, which shows a much stronger and significant negative effect.

5.2. Anatomy of the Response

To disentangle the profit response, we begin by examining how firms adjust their revenue and costs in reaction to marginal tax changes. Table 4 presents the elasticity estimates of reported profits—the same as our preferred estimates in Table 3 for comparability—alongside the elasticity estimates for reported revenue and costs, all with respect to the net-of-tax rate.

[Table 4 here]

The estimate based on the full sample indicates a positive reported profit elasticity in response to increases in the marginal net-of-tax rate of 3.9%, as shown in Column 1. This effect is primarily driven by a stronger response on the reported revenue side, with an estimated elasticity of 0.7%, compared to the reported costs side, which shows an elasticity of 0.5%, as seen in Columns 4 and 7, respectively. To better understand this result, we now separately examine the responsiveness of firms that experienced a

decrease in their *MTR* and those that faced an increase.

For firms that experienced a decrease in their *MTR*, the estimated elasticity is 4.5%, driven by positive responses in both reported revenue and costs. A decrease in the marginal tax rate—equivalently, an increase in the marginal net-of-tax rate—resulted in higher reported profits, explained by increases in both reported revenue and costs. However, the response in reported revenue was larger in magnitude (elasticity of 0.96%) than the response in reported costs (elasticity of 0.8%), which accounts for the overall net increase in reported profits.

One potential explanation for this is that the reform encouraged reported revenue growth, particularly for smaller firms that no longer needed to constrain their reported revenue to avoid exceeding the eliminated revenue threshold, T_1 , and thereby previously face a higher tax rate. The relevant group of firms that could be driving this result is G1 in Figure 6, referring to smaller firms that benefited from a reduced *MTR*. Results presented later in Subsection 5.3 support this hypothesis, as we show that the increase in reported profits was larger for Group G1 of smaller firms that benefited from the removal of these growth barriers, compared to Group G5, which consists of medium-sized firms that experienced a decrease in *MTR* but were not affected by this change (as the higher reported revenue threshold T_2 remained in place). Furthermore, the significant increases in employment and wagebill for firms that experienced a decrease in their *MTR*, as shown in Panels A and C of Appendix A2, suggest that at least part of these reported revenue and profit responses were real responses, though we cannot rule out the possibility that some were purely reporting responses. The fact that the response in reported costs was lower than that in reported revenues also suggests that the increases in reported revenues and profits were a combination of both real and purely reporting behaviour.

For firms that experienced an increase in the *MTR*, the estimated reported profit elasticity is 1.7%, indicating that a higher tax rate—equivalently, a lower marginal net-of-tax rate—leads to a decline in reported profits. In this case, both reported revenue and cost elasticity estimates are negative, at -1.9% and -3.2%, respectively. This suggests that while an increase in the *MTR* (a decrease in the marginal net-of-tax rate) led to higher reported revenue and costs, the larger magnitude of the cost increase resulted in a net decline in reported profits. One likely explanation is that this treated group

consists of small firms with relatively high profits (corresponding to Groups G3 and G4 in Figure 6), located near the reported revenue-dependent notch eliminated with the reform, T_1 . As indicated earlier, the removal of notch T_1 may have unlocked growth opportunities. The expansion of employment and wage bills documented in Panels B and D of Appendix A2 confirms that firms experiencing an increase in the *MTR* did experience a real expansion in their production. Consequently, these firms expanded their revenue reports, but the tax rate increase triggered an even stronger response in cost adjustments, ultimately leading to lower reported profits.

The larger magnitude of the reported costs response for firms with an increased *MTR* is also consistent with the findings of Bachas and Soto (2021), who explore Costa Rica's firms' responsiveness to higher tax rates in a static setting. They find that firms adjust more on the reported costs side than on the revenue side when facing higher tax rates, providing compelling evidence that these adjustments largely stem from tax evasion rather than real economic responses. The argument is that costs are easier to manipulate than revenue, making them a preferred channel for firms seeking to minimise tax liabilities. Moreover, the fact that the reform led to reported revenue growth in both groups of firms suggests that eliminating the revenue threshold T_1 removed distortions that had previously held back firms' expansion.

5.3. The Role of Simultaneous Variation in the *ATR*

We now investigate whether changes in the *ATR*, alongside those in the *MTR*, influenced firms' reported profit responses. Standard economic theory posits that, under rational after-tax profit-maximising behaviour, firms' intensive margin outcomes (such as reported profits) should respond to changes in the *MTR* rather than the *ATR*. However, empirical evidence suggests that consumers and individual taxpayers deviate from this standard theory when faced with relatively complex non-linear tax schedules. Instead, they sometimes react to average taxes or prices, a behaviour attributed to difficulties in fully understanding and oversimplifying the tax or price structures, a failure to react to the relevant incentives, leading to suboptimal decision-making (De Bartolome 1995; Ho 2023; Ito 2014; Rees-Jones and Taubinsky 2020; Shaffer 2020). Here, we examine whether this phenomenon also occurs among firms, for which there is currently no empirical evidence.

To investigate how firms react to variation in *MTR* and *ATR* in our context, we analyse the responses of each of the small and medium sized firms' groups defined in Figure 6 that experienced differently signed changes in both tax rates or at least in the *ATR* (G1 to G6), comparing them with the control group (GC) of only large firms, which saw no changes in either tax rate. As discussed in Section 4.1, firms experiencing a decrease in their *MTR* are expected to increase their reported profits, those experiencing an increase in their *MTR* should see a decrease, and those with no change in their *MTR* should not exhibit any variation in their reported profits. The predicted effects on reported profits for the different variabilities in *MTR* and *ATR* experienced by the seven groups are summarised in Table 5.

[Table 5 here]

Panel A of Table 6 presents the binary DiD analysis for the six treated groups. Both Group G1 (small firms) and Group G5 (medium firms) experienced decreases in both their *MTR* and *ATR*, leading, as expected, to an increase in reported profits following the reform, as shown in Columns 1 and 5. The magnitude of this effect appears slightly larger for G1, consisting of smaller firms. In contrast, Group G4 shown in Column 5, which faced increases in both rates, saw a decline in reported profits, as anticipated. Group G3 experienced an increase in *MTR* but a reduction in *ATR* which could have influenced firms' profit reporting behaviour. However, as indicated in Column 3, G3 followed the theoretical prediction by reducing reported profits, although in a lesser magnitude than G2.

[Table 6 here]

Next, we examine Groups G2 (small firms) and G6 (medium firms), both of which saw no change in *MTR* but a reduction in *ATR*. Theoretically, no significant change in their reported profits would be expected. Group G2's results confirm this prediction: the coefficient for being treated with only a decrease in *ATR* (and no change in *MTR*) is insignificant. However, Group G6 exhibits a significant and negative effect of being treated with a decrease in *ATR*, despite no change in *MTR*. Although this deviates from the theoretical predictions, we rule out the possibility that firms are responding to the decrease in *ATR*, as this would have led to an increase in reported profits. We hypothesise that this observed response may be driven by a subset of firms within G6 that

are near the second revenue threshold —medium-sized firms with relatively higher revenues that might be strategically managing their reported revenue to avoid crossing into the large-firm category. Even if the tax rate faced if passing the second revenue threshold has not changed, the opportunity cost of doing so may have gotten larger, particularly if their profits are below the profit threshold t_3 , as after the reform they could be dealing with a lower *MTR* of between 5% and 15%. The dynamics of these results are illustrated in Figure 9, obtained via the event-study DiD specifications. This figure confirms the patterns analysed above along a quick response to the tax changes.

[Figure 9 here]

Panel B of Table 6 presents the reported profit elasticity estimates in response to changes in the marginal net-of-tax rate for the groups that experienced a change in their *MTR* (G1, G3, G4, and G5). Given our treatment definition, groups G2 and G6 did not experience a change in their mechanical marginal net-of-tax rate, therefore, we cannot compute these estimates for them. These results reflect the asymmetries discussed in Subsection 5.1.3. Firms that faced a decrease in their *MTR* (G1 and G5) exhibit a stronger response, with elasticities of 9.92% and 3.98%, respectively, particularly stronger among smaller firms in group G1. In contrast, firms experiencing an increase in their *MTR* (G3 and G4) show a more lower response, with elasticities of 2.02% and 2.21%, respectively.

Finally, Panel C shows the reported profit elasticity estimates in response to changes in the marginal and average net-of-tax rates. Again, given the treatment definition, some groups do not have elasticity estimates for the marginal or average net-of-tax rates. Most treatment groups that experienced a change in their *ATRs* (G2, G3, G4 and G6), according to the expected no responsiveness of reported profits to changes in *ATR*, do not have significant estimates for the elasticities with respect to the average net-of-tax rate. The exception to this is G5, where the elasticity with respect to the average net-of-tax rate is significant, but negative: the response to the *ATR* seems to be attenuating the effects of the changes in the marginal net-of-tax rate ⁸.

These findings indicate no evidence that the reform led firms in Costa Rica to deviate from after-tax profit-maximising behaviour, despite operating under a relatively

⁸The strategy for simultaneously estimating the elasticities to the marginal and average net-of-tax rates is currently being refined

more complex CIT schedule and multiple changes in tax design. This result contrast with previous empirical evidence showing that other types of economic agents tend to respond to average tax rates, even when models of optimal behaviour predict responses to marginal taxes. Moreover, firms do not exhibit suboptimal responses, despite such behaviours being documented in developing economies with simpler tax structures.

We propose two potential, non-exclusive explanations for these findings. First, firms are generally considered more sophisticated economic agents and may therefore face fewer challenges or costs when responding to complex incentives. For example, they are more likely to have access to tax preparers and accountants, reducing the risk of misinterpretations. Additionally, firms in Costa Rica may be relatively more financially literate given the institutional framework, or have greater access to professional tax support, than those in countries where deviations from optimality have been observed under simpler tax schedules, such as Uganda and Rwanda (Almunia et al. 2024; Tourek 2022). Second, the details of the reform were widely publicised, and information was readily available on the *Ministerio de Hacienda* (Ministry of Finance) website—where the tax authority information is hosted and where tax declarations are made—, potentially reducing information frictions. Public interest in the reform was also evident, as suggested by a surge in Google searches for the term *Ministerio de Hacienda* around the announcement and implementation period (see Appendix A3). This suggests that well-communicated tax reforms—such as Costa Rica’s CIT reform—can help tax authorities implement less distortive systems while minimising errors and adjustment costs, even in the presence of added complexity.

5.3.1. Identification Checks

There are key conditions that must hold for the DiD strategy to be valid: the absence of anticipation effects, the parallel trends assumption, and the no-mean reversion condition. In this section, we perform the corresponding identification checks for our preferred specifications using the balanced panel.

No-anticipation. We first test the no-anticipation assumption by conducting placebo tests, estimating the event-study DiD specification while setting 2018 and 2019 fiscal years as hypothetical treatment start years instead of the 2020 fiscal year (which started in October 2019). As shown in Figure A4, there is no significant effect in either of the treatment groups for 2018 or 2019, while significant effects emerge in later years. Although

the reform was announced at the end of 2018, firms do not appear to have adjusted their behaviour before its enactment in the 2020 fiscal year. In addition, our treatment variables are constructed using tax reports from 2016 to 2019, covering three years of tax declarations filed before the new tax schedule was known. This helps mitigate concerns about anticipation effects and self-selection into treatment or control groups.

Parallel trends. We then assess parallel trends by visually comparing the evolution of profits between the treatment and control groups before the reform. Figure 8 shows that both groups followed similar trajectories prior to the policy change. This is also evident in Figure 9, which plots trends for treatment groups defined by different changes in *MTR* and *ATR*. In addition, Figure A5 presents event study coefficients from a standard specification without treatment interactions, showing the evolution of reported profits for the control and treatment groups from 2016 to 2023. This further confirms that trends were comparable before the reform.

No-mean reversion. Since the tax variation studied in this paper is correlated with pre-reform declared revenues and profits, it may be subject to the identification concern of mean reversion. As suggested in the literature (Saez et al. 2012), we test for mean reversion by including lags of our dependent variable, declared profits, as well as splines of pre-reform declared profits. Panel B in Table A4 shows that our results are robust to the inclusion of both the lag and spline terms, both qualitatively and quantitatively (see Panel A for our preferred results).

5.4. Robustness Checks

This subsection presents additional robustness tests conducted for our paper, which include testing for potential confounding effects related to COVID-19 and other policies introduced at the same time.

COVID-19. One of the most significant challenges of this paper is that the reform was introduced in the fiscal year 2020 (starting in October 2019), overlapping with the health crisis and the sanitary measures that restricted mobility and in-person activities from March 11, 2020, in Costa Rica. As discussed earlier, our preferred specifications include year and industry-year fixed effects to account for sector-specific impacts of the pandemic. Here, we provide further tests to support that our results are largely driven by the reform. First, we run a triple-difference specification by adding an interaction

term between the treatment effect and a dummy indicating whether a firm belongs to one of the three industry groups most affected by the COVID-19 shock. We define these industries as those with a GDP activity drop of more than 10% between 2019 and 2020: accommodation & food services (-36.3%), other services & NA (-18.1%), and wholesale, retail, transportation & storage (-10.8%)⁹. Panel B in Table A6 shows that our results remain robust compared to the main specifications in Panel A, and the triple-difference coefficient capturing the interaction between treatment and the most affected industries is not significant. Second, as an additional check, we exclude the two years most affected by the pandemic, 2020 and 2021 fiscal years, from our estimations. Figures A6 and A7 show that the results remain qualitatively consistent with our preferred estimates, which is also confirmed in Panel C of Table A6.

Simultaneous policies. Further robustness tests, currently *work-in-progress*, will explore the potential effects and interactions of other components of the Law on the Strengthening of Public Finances, which also introduced the CIT reform analysed here. In particular, we focus on the value added tax (VAT) reform, which replaced the previous sales tax system with a VAT. This reform changed both the rates and the structure of consumption taxation, with varying impacts across industries. It is possible that the VAT reform influenced firms' responses to the CIT reform, either by altering their overall tax burden or by affecting sector-specific conditions. Additional tests will help account for these effects and ensure that the responses identified are not confounded by overlapping policy changes.

5.5. Extensive Margin Analysis

Our analysis so far has focused on the intensive margin responses of firms to the reform. We are now turning to the extensive margin implications, including the effects on firm entry, exit, reporting positive revenues, and filing tax returns. This analysis is currently *work-in-progress*.

6. Conclusion

This paper examines how firms responded to Costa Rica's 2019 corporate income tax reform, which replaced a simple but distortionary notched tax schedule with a more complex kinked structure. The reform introduced substantial variation in firms' marginal

⁹See Table A5 for 2020 GDP growth data by industry

and average tax rates, generating both tax increases and reductions across similar firm types. Using administrative tax data from 2016 to 2023 and a difference-in-differences design, we document clear and rapid adjustments to the new tax schedule, in line with after-tax profit maximisation.

We uncover two main patterns. First, reported profits responded strongly to changes in marginal tax rates, but showed no sensitivity to changes in average tax rates. This is consistent with theoretical predictions, but contrasts with evidence from individuals who often simplify non-linear schedules and react to average rather than marginal incentives. Our findings suggest that firms in a middle-income setting with limited enforcement capacity can navigate more complex tax structures without misinterpreting the relevant tax incentives, especially when reforms are well-publicised. Second, we document an asymmetric response: firms reacted more to tax cuts than to tax increases. This asymmetry could reflect the elimination of growth barriers for small firms, particularly those that were previously constrained by the first revenue threshold, or the role of evasion costs, which may make reducing reported profits more costly and riskier in terms of audit exposure. While both mechanisms may play a role, our evidence suggests the latter may be more plausible, though further analysis is needed to confirm this.

Additionally, our descriptive analysis shows that the reform led to a significant decline in previously observed tax manipulation behaviours, such as bunching around revenue thresholds and profit margin discontinuities. This suggests that the reform was successful in reducing distortions, even with the added complexity. Overall, these findings highlight that more complex tax designs can be implemented effectively in middle-income countries, provided that reforms are well-communicated and accompanied by clarity in their implementation. Next steps will focus on extensive margin responses, the role of overlapping policies, and broader welfare implications of the reform with the support of a theoretical framework.

Tables

TABLE 1. Summary CIT Before and After the Reform

	Small and Medium Firms		Large Firms	
	Before	After	Before	After
Tax Base	Profits	Profits	Profits	Profits
Tax Rate function of	Revenue	Profits	Revenue	Revenue
Design	Notch	Kink	Notch	Notch
Tax Rates (%)	10, 20 (ATR)	5, 10, 15, 20 (MTR)	30 (ATR)	30 (ATR)

Notes: This table summarises the key features of Costa Rica's CIT system before and after the 2019 reform. The pre-reform system applied a notched design, where the average tax rate on reported profits was determined by reported revenue thresholds. Small and medium firms faced lower average tax rates of 10% and 20%, while large firms faced the statutory 30% average tax rate. After the reform, small and medium firms moved to a kinked schedule where the marginal tax rate applies to profits above certain thresholds, with rates of 5%, 10%, 15%, and 20%. Large firms continued under the same 30% ATR, with no changes in their tax schedule.

TABLE 2. Descriptive Statistics

	Small	Medium	Large	Total
Revenue (average, million CRC)	33.76	82.16	174.29	98.10
Costs (average, million CRC)	29.60	74.44	164.49	90.90
Profit margin (average, proportion)	0.15	0.11	0.07	0.11
Employment (average, number of employees)	2.13	3.57	6.32	4.06
Wagebill (average, million CRC)	7.54	14.35	27.68	16.80
Industry (%)				
<i>Agriculture, forestry & fishing</i>	9.59	6.16	6.17	7.38
<i>Manufacturing, mining & other industries</i>	8.15	10.09	10.57	9.57
<i>Construction</i>	3.68	5.44	5.26	4.74
<i>Wholesale, retail, transportation & storage</i>	23.3	29.12	37.03	29.89
<i>Accommodation & food services</i>	5.07	4.78	6.31	5.44
<i>Information & communication</i>	2.51	2.91	2.63	2.66
<i>Financial & insurance</i>	2.61	1.44	1.82	1.99
<i>Real estate</i>	8.45	6.77	3.34	6.13
<i>Prof, scientific, technical & admin</i>	16.44	15.46	14.5	15.46
<i>Education, human health & social work</i>	10.21	9.72	7.49	9.09
<i>Other services & NA</i>	10.01	8.12	4.89	7.63
N	32,208	25,608	32,560	90,376

Notes: This table presents descriptive statistics for the balanced panel sample used in our preferred estimations for the period 2016–2023. The sample consists of firms with non-missing employment data. Revenue, costs, and wage bill values are expressed in 2014 CRC. Firm size is determined based on pre-reform average reported revenue and profits. The profit margin is defined as the ratio of profits to revenues. Employment is measured as the average number of formal employees per month reported by firms to the Social Security Authority each year.

TABLE 3. Marginal Tax Rates Effects on Reported Profits

	(1) ↓ MTR+ ↑ MTR	(2) ↓ MTR	(3) ↑ MTR	(4) ↓ MTR+ ↑ MTR	(5) ↓ MTR	(6) ↑ MTR	(7) ↓ MTR+ ↑ MTR	(8) ↓ MTR	(9) ↑ MTR
PANEL A: DiD									
<i>D</i>	0.3608*** (0.0136)	0.6815*** (0.0140)	-0.1573*** (0.0180)	0.3619*** (0.0226)	0.5599*** (0.0233)	-0.1452*** (0.0362)	0.3122*** (0.0243)	0.4776*** (0.0248)	-0.1413*** (0.0396)
Implied ITT	0.4345	0.9768	-0.1456	0.3036	0.4287	-0.1351	0.3664	0.6122	-0.1318
Treated baseline average	3.97	2.27	15.62	4.53	3.29	14.31	4.53	3.29	14.31
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
N	315,321	287,492	129,038	90,376	84,952	47,480	90,376	84,952	47,480
PANEL B: Elasticity									
ε_{1-MTR}	4.7336*** (0.0979)	6.4599*** (0.1456)	1.9500*** (0.1979)	4.3830*** (0.1861)	5.3684*** (0.2243)	1.8181*** (0.4144)	3.9183*** (0.1821)	4.4867*** (0.2263)	1.7235*** (0.4541)
Treated baseline average	3.97	2.27	15.62	4.53	3.29	14.31	4.53	3.29	14.31
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	No	Yes	Yes	Yes
N	315,321	287,492	129,038	90,376	84,952	47,480	90,376	84,952	47,480

Notes: This table presents difference-in-differences estimations of the effects of changes in the marginal tax rate on firms' reported profits for the treatment groups (all treated firms, those experiencing a decrease in the marginal tax rate, and those experiencing an increase) relative to the control group. For all estimations, we determine treatment using the mechanical marginal tax rate change—calculated based on pre-reform revenue and reported profit information—and employ a Poisson quasi-maximum likelihood model to account for zeros in the dependent variable. All estimations incorporate individual and time fixed effects, with standard errors (in parentheses) clustered at the firm level. Columns 1–3 present results from an unbalanced-panel sample without additional controls. Columns 4–6 show results using a balanced-panel sample without additional controls, while Columns 7–9 display results from the balanced panel with additional controls. **These controls include year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins.** Panel A reports results from standard dichotomous difference-in-differences estimations. The first row presents the estimates directly from the regressions, while the third row provides the proportional effect, corresponding to an interpretable intention-to-treat (ITT) estimate. Panel B presents elasticity estimates using a continuous version of the treatment.

TABLE 4. Elasticities

	(1) ↓ <i>MTR</i> + ↑ <i>MTR</i> on Profits	(2) ↓ <i>MTR</i> on Profits	(3) ↑ <i>MTR</i> on Profits	(4) ↓ <i>MTR</i> + ↑ <i>MTR</i> on Revenue	(5) ↓ <i>MTR</i> on Revenue	(6) ↑ <i>MTR</i> on Revenue	(7) ↓ <i>MTR</i> + ↑ <i>MTR</i> on Costs	(8) ↓ <i>MTR</i> on Costs	(9) ↑ <i>MTR</i> on Costs
ϵ_{1-MTR}	3.9183*** (0.1821)	4.4867*** (0.2263)	1.7235*** (0.4541)	0.7208*** (0.0933)	0.9606*** (0.1017)	-1.8638*** (0.3189)	0.5259*** (0.0980)	0.8028*** (0.1034)	-3.2298*** (0.3871)
Treated baseline average	4.53	3.29	14.31	53.29	54.76	49.66	49.66	52.44	27.62
Balanced panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	90,376	84,952	47,480	90,376	84,952	47,480	90,376	84,952	47,480

Notes: This table presents elasticity estimates of profits (censored at zero), revenues, and costs with respect to the mechanical marginal tax rate, using a continuous version of the treatment. Estimates are reported by treatment group (all treated firms, those experiencing a decrease in the marginal tax rate, and those experiencing an increase) relative to the control group. Treatment is defined based on the mechanical marginal tax rate change, calculated using pre-reform revenue and reported profit information. Estimations employ a Poisson quasi-maximum likelihood model to account for zeros in the dependent variable and incorporate individual and time fixed effects, as well as year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors (in parentheses) are clustered at the firm level.

TABLE 5. Predicted Effects on Reported Profits by Treatment Groups

Group	Firm Size	ΔMTR	ΔATR	Predicted Effect on Reported Profits
G1	Small	↓	↓	↑
G2	Small	=	↓	=
G3	Small	↑	↓	↓
G4	Small	↑	↑	↓
G5	Medium	↓	↓	↑
G6	Medium	=	↓	=
GC	Large	=	=	=

Notes: This table presents the predicted effects of the tax change on reported profits for each of the six treatment groups, defined based on changes in their marginal and average tax rates.

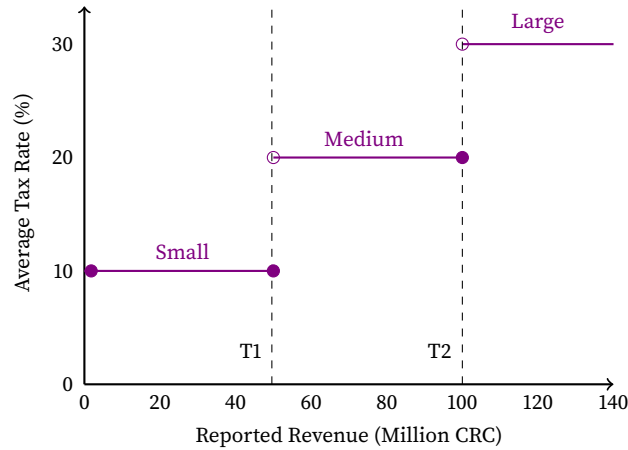
TABLE 6. Marginal and Average Tax Rates Effects on Reported Profits

	(1) Group G1 ↓ MTR, ↓ ATR	(2) Group G2 = MTR, ↓ ATR	(3) Group G3 ↑ MTR, ↓ ATR	(4) Group G4 ↑ MTR, ↑ ATR	(5) Group G5 ↓ MTR, ↓ ATR	(6) Group G6 = MTR, ↓ ATR
PANEL A: DiD						
<i>D</i>	0.4962*** (0.0444)	-0.0293 (0.0605)	-0.1223** (0.0576)	-0.2209*** (0.0602)	0.4061*** (0.0323)	-0.1127*** (0.0404)
Implied ITT	0.6425	-0.0289	-0.1151	-0.1982	0.5010	-0.1066
Treated baseline average	2.21	6.76	10.33	18.60	4.52	21.10
Balanced panel	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes
N	55,488	36,416	35,368	35,176	52,528	38,200
PANEL B: Elasticity						
ϵ_{1-MTR}	9.9234*** (0.8889)	NA (.)	2.0169** (0.7879)	2.2088*** (0.6023)	3.9838*** (0.2586)	NA (.)
Treated baseline average	2.21	NA	10.33	18.60	4.52	NA
Balanced panel	Yes	NA	Yes	Yes	Yes	NA
Additional controls	Yes	NA	Yes	Yes	Yes	NA
N	55,488	NA	35,368	35,176	52,528	NA
PANEL C: Elasticities						
ϵ_{1-MTR}	9.9234*** (0.8889)	NA (.)	1.8767** (0.9118)	2.0349** (0.9841)	6.1002*** (0.7382)	NA (.)
ϵ_{1-ATR}	NA (.)	-0.5732 (1.4524)	-0.7709 (3.0995)	0.5177 (2.4350)	-1.9343*** (0.6480)	-0.6100 (0.4994)
Treated baseline average	2.21	6.76	10.33	18.60	4.52	21.10
Balanced panel	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes
N	55,488	36,416	35,368	35,176	52,528	38,200

Notes: This table presents difference-in-differences estimations of the effects of changes in marginal and average tax rates on firms' reported profits for the six treatment groups, defined based on changes in these tax rates. Treatment is determined using the mechanical marginal average tax rate changes, calculated from pre-reform revenue and reported profit information. A Poisson quasi-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate individual and time fixed effects, as well as year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors (in parentheses) are clustered at the firm level. Panel A reports results from standard dichotomous difference-in-differences estimations, with the first row presenting estimates directly from the regressions and the third row providing the proportional effect, corresponding to an interpretable intention-to-treat (ITT) estimate. Panel B presents elasticity estimates using a continuous version of the marginal tax rate treatment. Panel C presents elasticity estimates using continuous versions of both the marginal and average tax rate treatments.

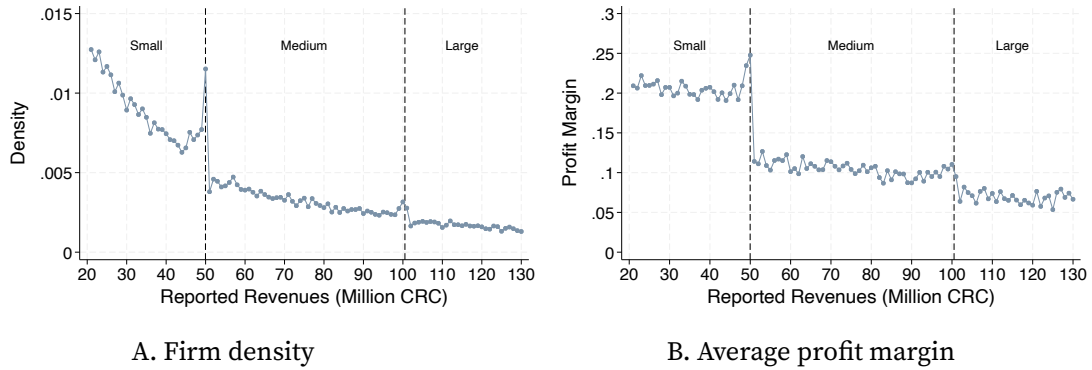
Figures

FIGURE 1. Corporate Tax Schedule Before the Reform



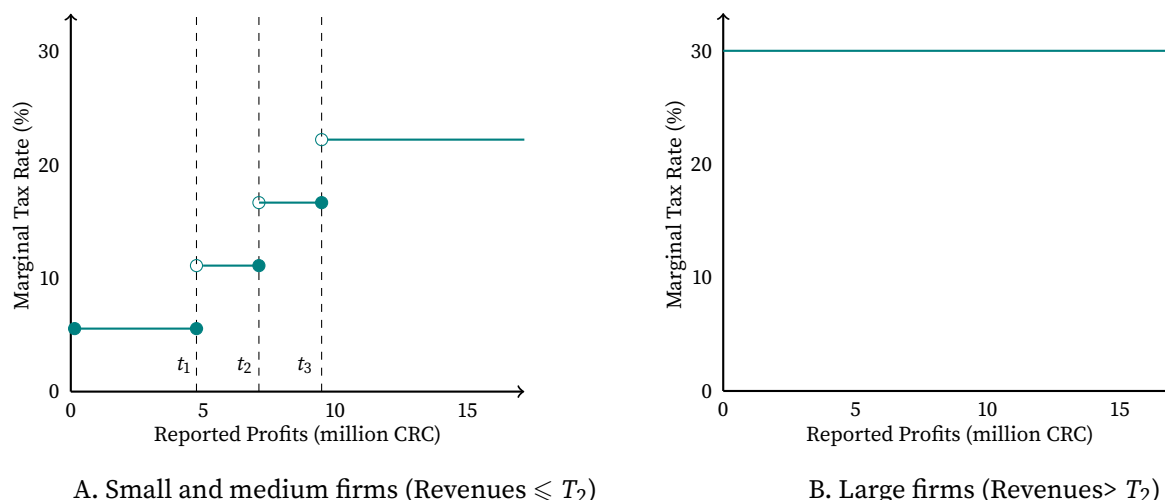
Notes: This figure shows the notched corporate tax schedule faced by all firms previous to the 2019 reform. Firms faced increasing average tax rates on their reported profits, which were a function of their revenues. *Small* firms with revenues up to the first threshold $T1$ faced an average tax rate of 10 percent, *medium* firms with revenues beyond the first threshold and up to the second threshold $T2$ faced a rate of 20 percent, and *large* firms with revenues beyond the second threshold faced an average tax rate of 30 percent.

FIGURE 2. Firm Density and Average Profit Margin Before the Reform



Notes: This figure shows the key patterns of the corporate tax data before the reform, specifically for the years 2016, 2017 and 2018. Panel A displays firm density by revenue bin. Panel B shows average profit margin by revenue bin, where profit margin is defined as profit over revenue. Bin size is 1 million CRC.

FIGURE 3. Corporate Tax Schedule After the Reform



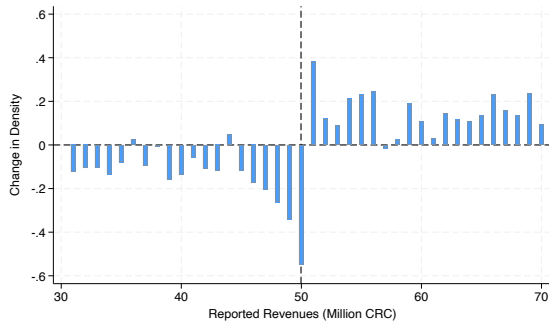
Notes: Panel A shows the kinked profit tax schedule faced by small and medium sized firms after the 2019 reform. Small and medium firms face increasing marginal tax rates on their reported profits, which are a function of their reported profits. Firms with reported profits up to the first threshold t_1 face a marginal tax rate of 5 percent, firms with reported profits beyond the first threshold and up to the second threshold t_2 face a rate of 10 percent, firms with reported profits beyond the second threshold and up to the third threshold t_3 face a rate of 15 percent, and firms with reported profits beyond the third threshold face an marginal tax rate of 20 percent. Panel B shows that large firms kept the same average and marginal tax rate on reported profits of 30 percent, here expressed by level of reported profits.

FIGURE 4. Firm Density, and Average Profit Margin, Costs and Employment, Before and After the Reform

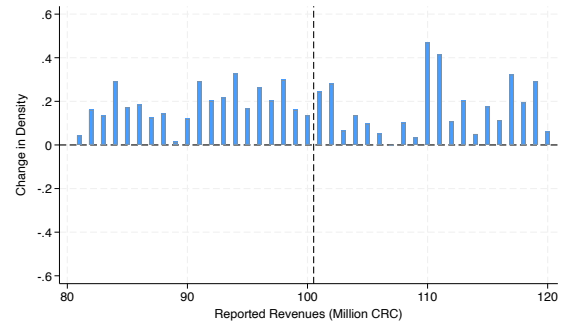


Notes: This figure shows the key patterns of the firms' data before and after the reform, for the years 2016-2018 in blue and 2021-2023 in orange, respectively. Panel A displays firm density by reported revenue bin. Panel B shows average profit margin by reported revenue bin, where profit margin is defined as profit over revenue. Panel C shows average reported costs by reported revenue bin. Panel D illustrates average number of employees registered by the firms to the Social Security Administration by reported revenue bin. Bin size is 1 million 2014 CRC.

FIGURE 5. Firm Density Changes



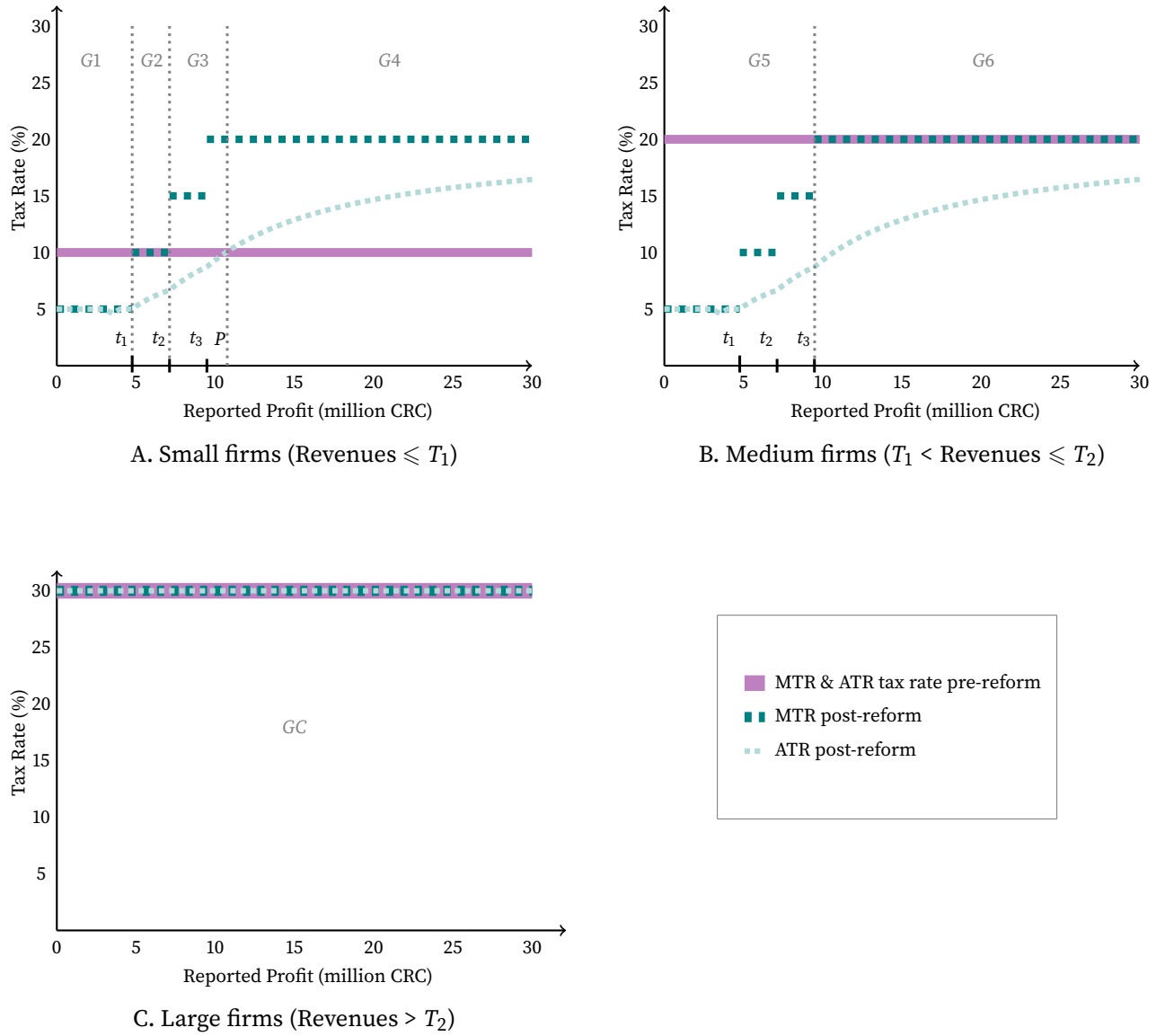
A. Around T1 threshold



B. Around T2 threshold

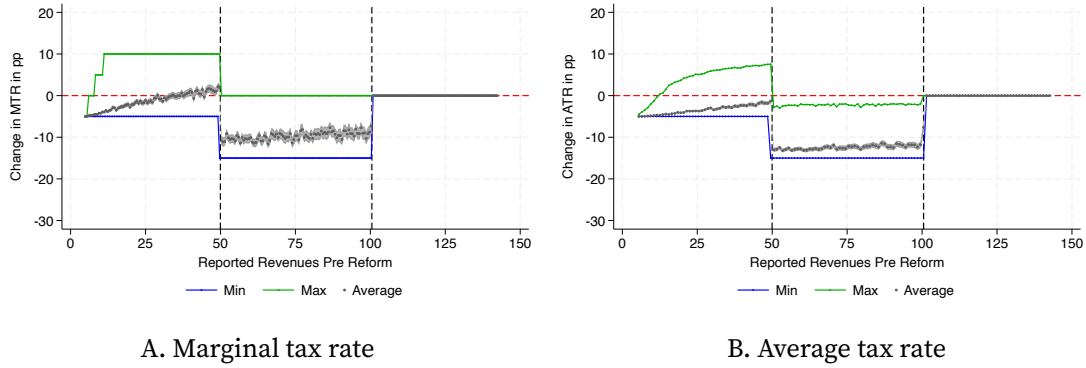
Notes: This figure illustrates changes in accumulated firm density by 1 million 2014 CRC bins of reported revenue, comparing the periods before (2016–2018) and after the reform (2021–2023). Panel A presents these changes around the first reported revenue threshold, while Panel B shows those around the second reported revenue threshold.

FIGURE 6. Marginal and Average Tax Rates, Before and After the Reform



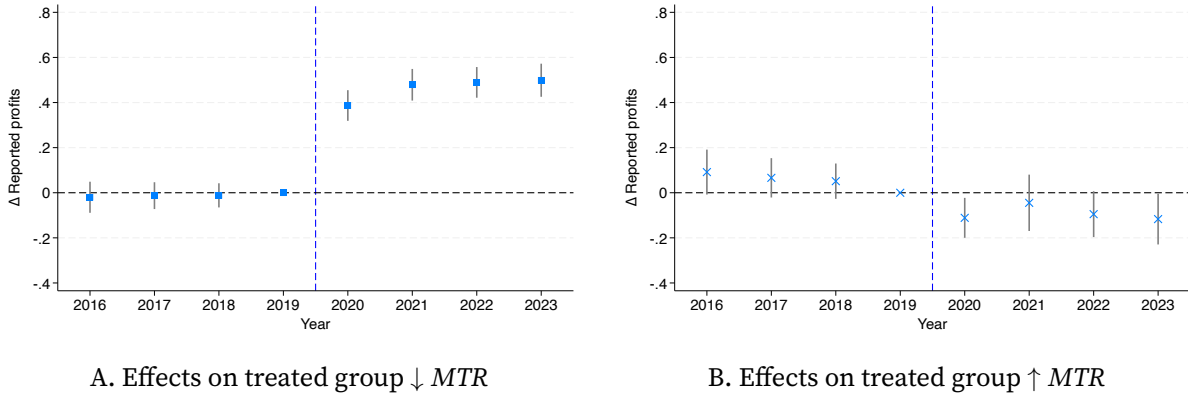
Notes: This figure illustrates the marginal and average tax rates faced by firms before and after the tax reform, assuming constant reported revenue and profits. Panel A shows that small firms faced a marginal and average tax rate of 10% before the reform. After the reform, they encounter increasing stepped marginal tax rates ranging from 5% to 20%, and a smooth average tax rate increasing from 5% to under 20%. Panel B reveals that medium-sized firms were subject to a marginal and average tax rate of 20% before the reform. Post-reform, these firms face increasing stepped marginal tax rates from 5% to 20%, and a smooth average tax rate increasing from 5% to below 20%. Panel C shows that large firms maintained a marginal and average tax rate of 30% both before and after the reform. Firms are categorized into one of seven groups based on the changes in their mechanical marginal and average tax rates (i.e., the changes they would have experienced due to the tax reform, assuming no alteration in the reported information). This categorisation depends on the firm's size and reported profit level. The first six groups, from G1 to G6, are treatment groups since they experienced a change in at least the average tax rate. The final group, GC, is the control group as it did not experience any change in the marginal or average mechanical tax rates.

FIGURE 7. Reform-Induced Mechanical Marginal and Average Tax Rate Changes



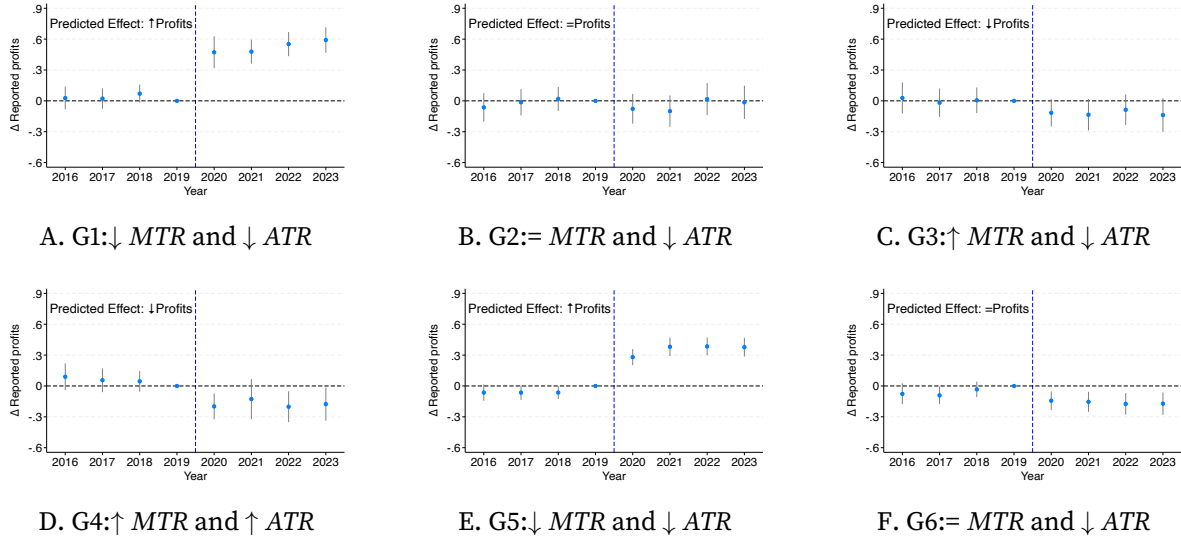
Notes: Panel A illustrates the average, minimum, and maximum mechanical marginal tax rate changes, while Panel B illustrates the average, minimum, and maximum mechanical average tax rate changes induced by the reform. Both are calculated based on firms' average reported revenue and profits from 2016–2019, by 1 million 2014 CRC reported revenue bins.

FIGURE 8. Difference-in-Differences by MTR Treatment Groups



Notes: This figure presents the 2016–2023 difference-in-differences event study estimates of the dynamic effects of the reform on reported profits for treatment groups that experienced changes in their marginal tax rates, compared to the control group, using our preferred balanced-panel sample. Treatment is determined using the mechanical marginal tax rate change, calculated from pre-reform revenue and reported profit information. A Poisson quasi-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate individual and time fixed effects, as well as year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors are clustered at the firm level. Panel A displays the effects for firms that experienced a decrease in the marginal tax rate, while Panel B shows the effects for those facing an increase.

FIGURE 9. Difference-in-Differences by MTR and ATR Treatment Groups



Notes: This figure presents the 2016-2023 difference-in-differences event study estimates of the dynamic effects of the reform on reported profits for the six treatment groups, defined based on changes in marginal and average tax rates, using our preferred balanced-panel sample. Treatment is determined using the mechanical changes in marginal and average tax rates, calculated from pre-reform revenue and reported profit information. A Poisson quasi-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate individual and time fixed effects, as well as year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors are clustered at the firm level. Panel A shows the effects for Group 1, which consists of small firms that experienced a decrease in both their marginal and average tax rates. Panel B displays the effects for Group 2, including small firms that experienced no change in their marginal tax rate but a decrease in their average tax rate. Panel C illustrates the effects for Group 3, comprising firms that experienced an increase in their marginal tax rate and a decrease in their average tax rate. Panel D presents the effects for Group 4, featuring firms that experienced an increase in both their marginal and average tax rates. Panel E presents the effects for Group 5, consisting of medium-sized firms that experienced an increase in both their marginal and average tax rates. Panel F shows the effects for Group 6, comprising medium-sized firms that experienced no change in their marginal tax rate but a decrease in their average tax rate.

7. Appendix

TABLE A1. Descriptive Statistics, Unalanced-Panel Sample

	Small	Medium	Large	Total
Revenue (average, million CRC)	29.68	82.79	178.59	74.08
Costs (average, million CRC)	25.47	75.40	169.67	68.16
Profit margin (average, proportion)	0.19	0.12	0.07	0.15
Industry (%)				
<i>Agriculture, forestry & fishing</i>	7.20	5.70	5.89	6.60
<i>Manufacturing, mining & other industries</i>	6.53	7.86	8.98	7.35
<i>Construction</i>	5.08	6.87	6.76	5.82
<i>Wholesale, retail, transportation & storage</i>	21.79	30.02	38.32	27.20
<i>Accommodation & food services</i>	3.23	4.23	6.42	4.15
<i>Information & communication</i>	3.65	3.04	2.57	3.28
<i>Financial & insurance</i>	2.71	1.75	1.88	2.33
<i>Real estate</i>	9.08	7.11	3.69	7.46
<i>Prof, scientific, technical & admin</i>	21.25	16.64	14.10	18.70
<i>Education, human health & social work</i>	7.35	8.03	6.12	7.21
<i>Other services & NA</i>	12.14	8.75	5.26	9.90
N	179,886	64,360	71,075	315,321

Notes: This table presents the descriptive statistics for the 2016-2023 unbalanced panel-sample. Values for revenue, costs and wagebill are expressed in 2014 CRC. Firm size determined according to reported revenue and profits pre-reform averages. The profit margin is defined as the ratio of profits to revenues. Employment is defined as the average number of formal employees per month reported by firms to the Social Security authority each year.

TABLE A2. Heterogeneity by Firm Size: Marginal Tax Rates Effects on Reported Profits

	(1) ↓ <i>MTR</i> + ↑ <i>MTR</i>	(2) ↓ <i>MTR</i>	(3) ↑ <i>MTR</i>	(4) ↓ <i>MTR</i> + ↑ <i>MTR</i>	(5) ↓ <i>MTR</i>	(6) ↑ <i>MTR</i>	(7) ↓ <i>MTR</i> + ↑ <i>MTR</i>	(8) ↓ <i>MTR</i>	(9) ↑ <i>MTR</i>
<i>D</i>	0.5406*** (0.0176)	0.8157*** (0.0181)	-0.1393*** (0.0284)	0.4289*** (0.0475)	0.6751*** (0.0473)	-0.4262*** (0.1050)	0.3632*** (0.0520)	0.5925*** (0.0526)	-0.3908*** (0.1109)
<i>D</i> * <i>Larger</i> small	-0.4126*** (0.0193)	-0.0856*** (0.0227)	-0.0232 (0.0314)	-0.2636*** (0.0516)	-0.0842 (0.0550)	0.3148*** (0.1094)	-0.2327*** (0.0526)	-0.0751 (0.0557)	0.2732** (0.1120)
<i>D</i> * <i>Smaller</i> medium	-0.0032 (0.0229)	-0.2784*** (0.0233)	NA (.)	0.0724 (0.0521)	-0.1739*** (0.0519)	NA (.)	0.0641 (0.0539)	-0.1641*** (0.0542)	NA (.)
<i>D</i> * <i>Larger</i> medium	0.0256 (0.0293)	-0.2496*** (0.0296)	NA (.)	0.1187* (0.0607)	-0.1275** (0.0605)	NA (.)	0.1075* (0.0641)	-0.1187* (0.0649)	NA (.)
Treated baseline average	3.97	2.27	15.62	4.53	3.29	14.31	4.53	3.29	14.31
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	No	Yes	Yes	Yes
N	315,321	287,492	129,038	90,376	84,952	47,480	90,376	84,952	47,480

Notes: This table presents triple-difference estimations of the effects of changes in the marginal tax rate on firms' reported profits, by firm size group. The treatment group includes firms that experienced a change in their mechanical marginal tax rate, calculated using pre-reform revenue and reported profit information. The main treatment effect (*D*) captures the effect for the baseline size category (*smaller* small firms). Interaction terms with firm size dummies indicate how this effect differs across the other size groups: *larger* small, *smaller* medium, and *larger* medium. The last two medium firm size categories are excluded in the specifications for the increase in the marginal tax rate, as only small firms experienced a decrease in their marginal tax rate. All estimations are performed using a Poisson quasi-maximum likelihood model to account for zeros in the dependent variable, and include individual and year fixed effects. Additional controls in columns 7–9 include year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors, in parentheses, are clustered at the firm level.

TABLE A3. Heterogeneity by Industry: Marginal Tax Rates Effects on Reported Profits

	(1) ↓ MTR + ↑ MTR	(2) ↓ MTR	(3) ↑ MTR	(4) ↓ MTR + ↑ MTR	(5) ↓ MTR	(6) ↑ MTR	(7) ↓ MTR + ↑ MTR	(8) ↓ MTR	(9) ↑ MTR
<i>D</i>	0.4293*** (0.0260)	0.7114*** (0.0320)	0.1339*** (0.0353)	0.3449*** (0.0431)	0.5107*** (0.0489)	0.0439 (0.0702)	0.1729*** (0.0563)	0.3265*** (0.0609)	-0.1308* (0.0788)
<i>D</i> * Agriculture, forestry & fishing	0.0736* (0.0442)	0.0629 (0.0469)	-0.4838*** (0.0884)	0.2358*** (0.0752)	0.2941*** (0.0765)	-0.3682** (0.1600)	0.4162*** (0.1075)	0.4696*** (0.1079)	-0.1730 (0.1786)
<i>D</i> * Manufacturing, mining & other industries	0.1099** (0.0444)	-0.0044 (0.0493)	-0.2244** (0.0933)	0.1288* (0.0686)	0.0730 (0.0700)	-0.1480 (0.2162)	0.2114** (0.0943)	0.1566 (0.0962)	-0.0633 (0.2190)
<i>D</i> * Construction	0.0264 (0.0458)	-0.0128 (0.0487)	-0.3266*** (0.0896)	0.2105** (0.0859)	0.1521 (0.0950)	0.1714 (0.1705)	0.2937** (0.1183)	0.2293* (0.1247)	0.2909 (0.1906)
<i>D</i> * Wholesale, retail, transportation & storage	0.0725** (0.0295)	0.0012 (0.0348)	-0.3459*** (0.0512)	0.0953* (0.0518)	0.0487 (0.0584)	-0.0465 (0.0937)	0.1485** (0.0686)	0.1008 (0.0739)	0.0136 (0.1037)
<i>D</i> * Accommodation & food services	0.0332 (0.0512)	-0.0771 (0.0547)	-0.3464*** (0.1180)	0.1004 (0.0912)	0.0520 (0.0960)	-0.2218 (0.2031)	0.2428** (0.1131)	0.1900 (0.1171)	-0.0778 (0.2144)
<i>D</i> * Information & communication	0.1057** (0.0456)	0.1041** (0.0522)	-0.1302* (0.0697)	0.2320*** (0.0878)	0.1637 (0.0997)	0.1778 (0.1355)	0.2030* (0.1218)	0.1242 (0.1307)	0.1713 (0.1674)
<i>D</i> * Financial & insurance	-0.3982*** (0.0577)	0.1410 (0.0990)	-0.3795*** (0.0674)	-0.5933*** (0.1430)	-0.1790 (0.2269)	-0.4701*** (0.1820)	0.0213 (0.1833)	0.4364* (0.2601)	0.1554 (0.2121)
<i>D</i> * Real estate	-0.3347*** (0.0373)	-0.1523*** (0.0476)	-0.3278*** (0.0525)	-0.2684*** (0.0757)	-0.1158 (0.0986)	-0.2290** (0.1125)	0.0728 (0.0986)	0.2052* (0.1147)	0.1226 (0.1285)
<i>D</i> * Prof, scientific, technical & admin	-0.1411*** (0.0291)	-0.0428 (0.0354)	-0.2913*** (0.0427)	-0.0968* (0.0536)	-0.0154 (0.0582)	-0.3636*** (0.0990)	0.0332 (0.0752)	0.1123 (0.0786)	-0.2267** (0.1118)
<i>D</i> * Other services & NA	-0.1768*** (0.0361)	-0.1767*** (0.0424)	-0.4828*** (0.0606)	-0.0278 (0.0721)	-0.0140 (0.0798)	-0.3288*** (0.1261)	0.1916* (0.1009)	0.2013* (0.1065)	-0.1148 (0.1435)
Treated baseline average	3.97	2.27	15.62	4.53	3.29	14.31	4.53	3.29	14.31
Balanced panel	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	No	Yes	Yes	Yes
N	315,321	287,492	129,038	90,376	84,952	47,480	90,376	84,952	47,480

Notes: This table presents triple-difference estimations of the effects of changes in the marginal tax rate on firms' reported profits, by industry. The treatment group includes firms that experienced a change in their mechanical marginal tax rate, calculated using pre-reform revenue and reported profit information. The main treatment effect (*D*) reflects the effect for the baseline industry category, defined as education, human health & social work. The interaction terms with industry dummies indicate how the treatment effect varies across the other industries. All estimations use a Poisson quasi-maximum likelihood model to account for zeros in the dependent variable and include individual and year fixed effects. Additional controls in columns 7–9 include year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors, in parentheses, are clustered at the firm level.

TABLE A4. Mean Reversion Robustness: Marginal Tax Rates Effects on Reported Profits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	$\downarrow MTR + \uparrow MTR$	$\downarrow MTR$	$\uparrow MTR$	Group G1 $\downarrow MTR, \downarrow ATR$	Group G2 $= MTR, \downarrow ATR$	Group G3 $\uparrow MTR, \downarrow ATR$	Group G4 $\uparrow MTR, \uparrow ATR$	Group G5 $\downarrow MTR, \downarrow ATR$	Group G6 $= MTR, \downarrow ATR$
PANEL A: DiD									
<i>D</i>	0.3122*** (0.0243)	0.4776*** (0.0248)	-0.1413*** (0.0396)	0.4962*** (0.0444)	-0.0293 (0.0605)	-0.1223** (0.0576)	-0.2209*** (0.0602)	0.4061*** (0.0323)	-0.1127*** (0.0404)
Treated baseline average	4.53	3.29	14.31	2.21	6.76	10.33	18.60	4.52	21.10
Balanced Panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes			
N	90,376	84,952	47,480	55,488	36,416	35,368	35,176	52,528	38,200
PANEL B: Mean reversion robustness									
<i>D</i> with lag profits	0.3045*** (0.0208)	0.4633*** (0.0219)	-0.1237*** (0.0326)	0.4864*** (0.0383)	-0.0415 (0.0525)	-0.1210** (0.0476)	-0.2122*** (0.0476)	0.3747*** (0.0281)	-0.1543*** (0.0314)
<i>D</i> with lag profits + splines	0.2565*** (0.0201)	0.3974*** (0.0213)	-0.0955*** (0.0324)	0.4126*** (0.0379)	-0.0597 (0.0491)	-0.0942** (0.0450)	-0.1671*** (0.0483)	0.3206*** (0.0271)	-0.1183*** (0.0310)
Treated baseline average	4.53	3.29	14.31	2.21	6.76	10.33	18.60	4.52	21.10
Balanced Panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	78,925	74,179	41,475	48,433	31,794	30,877	30,709	45,857	33,355

Notes: This table presents robustness tests for the difference-in-differences estimations of the effects of changes in the marginal tax rate on firms' reported profits, focusing on mean reversion concerns. Panel A replicates the preferred specifications from Table 3 and 6. Panel B adds a lag of declared profits as a control, while Panel C further includes splines of pre-reform declared profits. Including these terms reduces the number of observations compared to Panel A, as the inclusion of the lagged dependent variable leads to the loss of the 2016 observations. The treatment groups correspond to the three basic definitions (change in marginal tax rate, decrease in marginal tax rate, and increase in marginal tax rate) and the six alternative treatment group definitions described in the paper. All estimations use a Poisson quasi-maximum likelihood model, include individual and year fixed effects, and control for year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors, in parentheses, are clustered at the firm level.

TABLE A5. GDP 2019 and GDP Growth 2020, by Industry

	GDP 2019	GDP Growth 2020
Agriculture, forestry & fishing	1,654,172	-1.4
Manufacturing, mining & other industries	5,335,924	2.9
Construction	1,454,936	-0.9
Wholesale, retail, transportation & storage	4,869,581	-10.8
Accommodation & food services	1,208,238	-36.3
Information & communication	1,601,438	1.8
Financial & insurance	2,028,405	5.6
Real estate	2,938,306	-2.3
Prof, scientific, technical & admin	4,530,372	-3.2
Education, human health & social work	5,032,630	-1.8
Other services & NA	1,313,058	-18.1

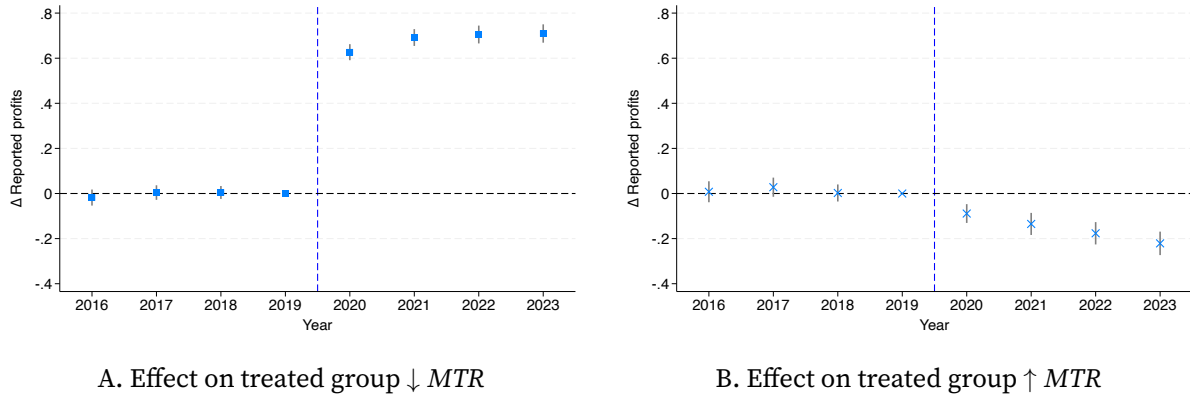
Notes: This table presents Costa Rica's 2019 GDP and 2020 GDP growth rates, both by industry, used to define the industries most affected by the COVID-19 shock. Industries with a GDP contraction in 2020 larger than 10%—specifically accommodation & food services; other services & NA; and wholesale, retail, transportation & storage— are classified as the most affected industries (in bold). This classification is used in the triple-difference specification to test the robustness of our main results to potential COVID-19 confounding effects.

TABLE A6. COVID-19 Robustness: Marginal Tax Rates Effects on Reported Profits

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	↓ MTR+ ↑ MTR	↓ MTR	↑ MTR	Group G1 ↓ MTR, ↓ ATR	Group G2 = MTR, ↓ ATR	Group G3 ↑ MTR, ↓ ATR	Group G4 ↑ MTR, ↑ ATR	Group G5 ↓ MTR, ↓ ATR	Group G6 = MTR, ↓ ATR
PANEL A: DiD									
<i>D</i>	0.3122*** (0.0243)	0.4776*** (0.0248)	-0.1413*** (0.0396)	0.4962*** (0.0444)	-0.0293 (0.0605)	-0.1223** (0.0576)	-0.2209*** (0.0602)	0.4061*** (0.0323)	-0.1127*** (0.0404)
Treated baseline average	4.53	3.29	14.31	2.21	6.76	10.33	18.60	4.52	21.10
Balanced Panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes			
N	90,376	84,952	47,480	55,488	36,416	35,368	35,176	52,528	38,200
PANEL B: DDD									
<i>D</i>	0.2929*** (0.0302)	0.4929*** (0.0307)	-0.1357*** (0.0489)	0.5267*** (0.0521)	-0.0428 (0.0711)	-0.1456** (0.0692)	-0.2071*** (0.0729)	0.3865*** (0.0404)	-0.1360*** (0.0489)
<i>D * COVID – 19 more affected</i>	0.0486 (0.0445)	-0.0354 (0.0463)	-0.0184 (0.0739)	-0.0739 (0.0642)	0.0379 (0.1031)	0.0636 (0.0968)	-0.0490 (0.1036)	0.0431 (0.0578)	0.0684 (0.0780)
Treated baseline average	4.53	3.29	14.31	2.21	6.76	10.33	18.60	4.52	21.10
Balanced Panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	90,376	84,952	47,480	55,488	36,416	35,368	35,176	52,528	38,200
PANEL B: DiD Removing 2020 and 2021									
<i>D</i>	0.3608*** (0.0136)	0.6815*** (0.0140)	-0.1573*** (0.0180)	0.4962*** (0.0444)	-0.0293 (0.0605)	-0.1223** (0.0576)	-0.2209*** (0.0602)	0.4061*** (0.0323)	-0.1127*** (0.0404)
Treated baseline average	4.53	3.29	14.31	2.21	6.76	10.33	18.60	4.52	21.10
Balanced Panel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	90,376	84,952	47,480	55,488	36,416	35,368	35,176	52,528	38,200

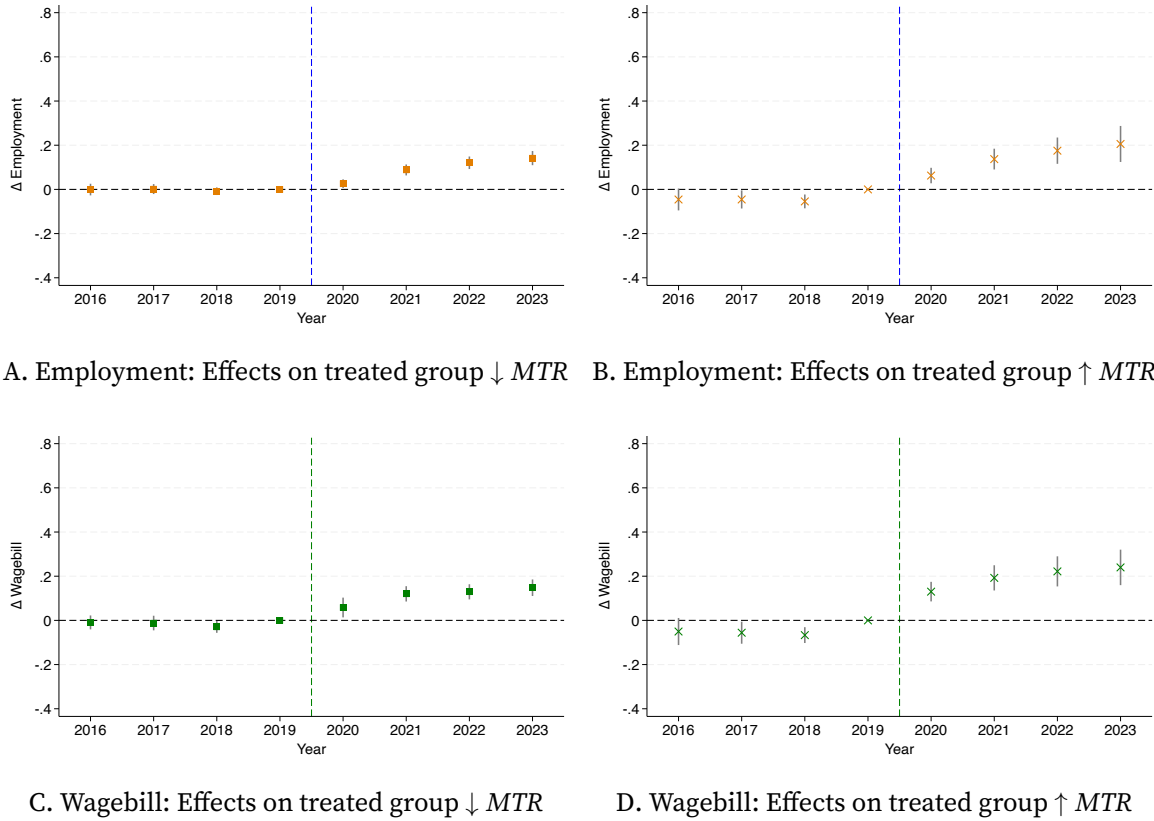
Notes: This table presents robustness checks for the difference-in-differences estimations of the effects of changes in the marginal tax rate on firms' reported profits, focusing on the potential confounding effects of COVID-19. Panel A replicates the preferred specifications from Table 3 and 6. Panel B includes a triple-difference term interacting the treatment variable with an indicator for firms in the industries most affected by COVID-19, as defined in Table A5. Panel C excludes the 2020 and 2021 fiscal years—those most affected by the pandemic—from the sample. The treatment groups correspond to the three basic definitions (change in marginal tax rate, decrease in marginal tax rate, and increase in marginal tax rate) and the six alternative treatment group definitions described in the paper. All estimations use the balanced panel, are based on Poisson quasi-maximum likelihood models, include individual and year fixed effects, and control for year fixed effects interacted with pre-reform industry classifications, average wage bill bins, and average wage bill-to-revenue ratio bins. Standard errors, in parentheses, are clustered at the firm level.

FIGURE A1. Difference-in-Differences by MTR Treatment Groups,
Unbalanced-Panel Sample



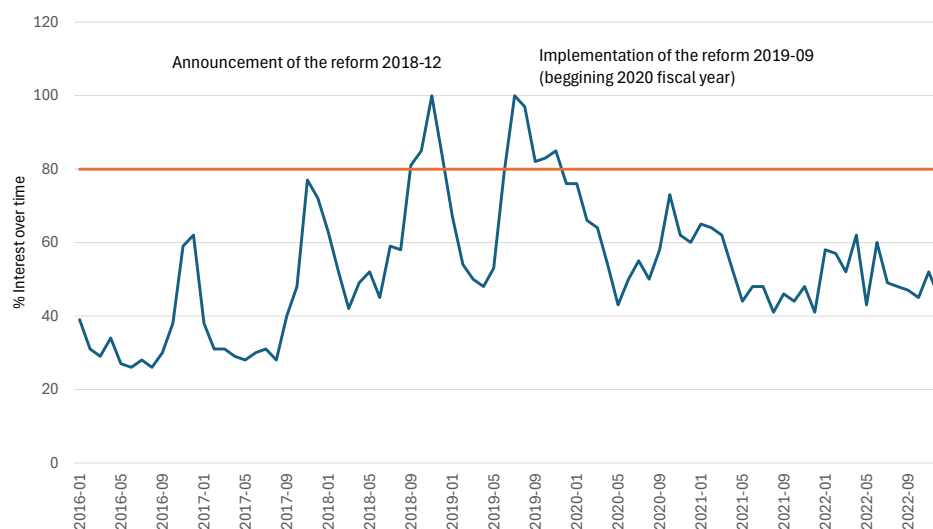
Notes: This figure presents the 2016-2023 difference-in-differences event study estimates of the dynamic effects of the reform on reported profits for treatment groups that experienced changes in their marginal tax rates, compared to the control group, using the unbalanced-panel sample. Treatment is determined using the mechanical marginal tax rate change, calculated from pre-reform revenue and reported profit information. A Poisson quasi-maximum likelihood model is employed to account for zeros in the dependent variable. All estimations incorporate individual and time fixed effects. Standard errors are clustered at the firm level. Panel A displays the effects for firms that experienced a decrease in the marginal tax rate, while Panel B shows the effects for those facing an increase.

FIGURE A2. Difference-in-Differences by MTR Treatment Groups,
Employment and Wagebill



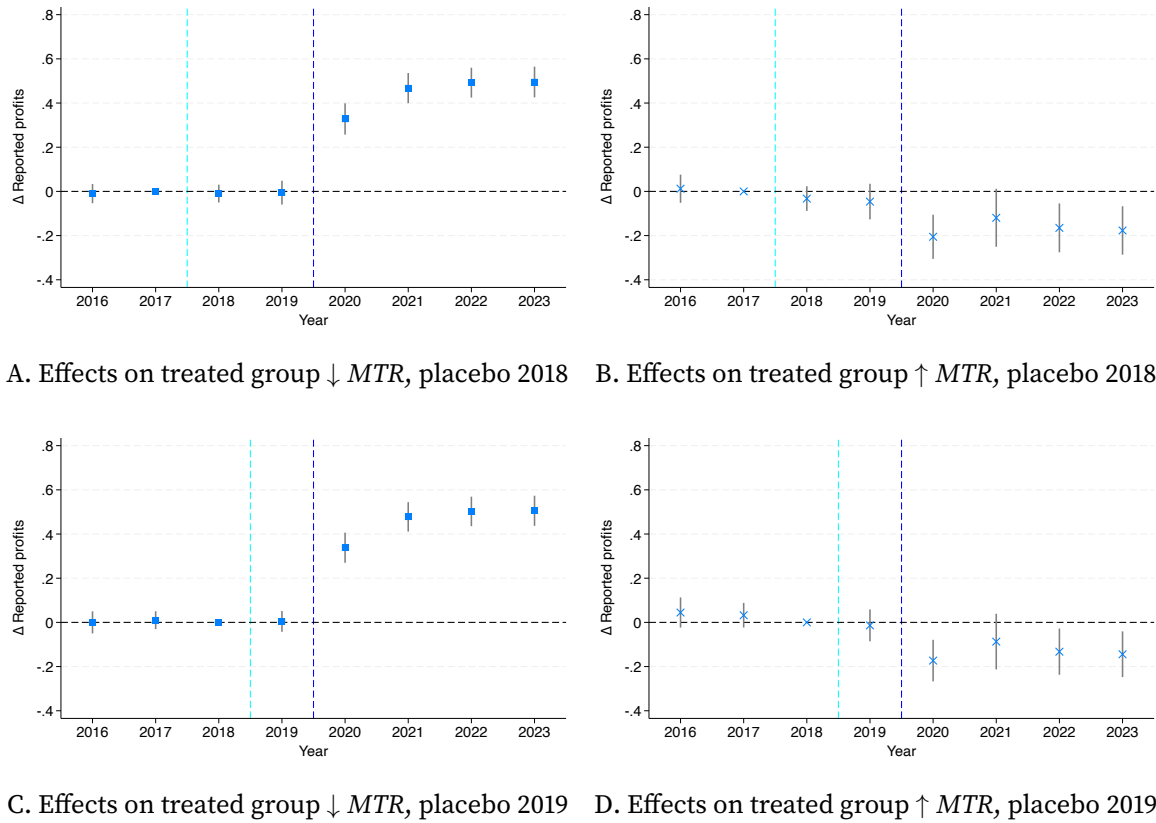
Notes: This figure presents the 2016–2023 difference-in-differences event study estimates of the dynamic effects of the reform on firms’ average monthly employment and wage bill for treatment groups that experienced changes in their marginal tax rates, compared to the control group, using our preferred balanced-panel sample. Treatment is determined based on the mechanical marginal tax rate change, calculated from pre-reform revenue and reported profit information. A Poisson quasi-maximum likelihood model is used to account for zeros in the dependent variable. All estimations include individual and time fixed effects. Standard errors are clustered at the firm level. Panels A and B show the effects on employment for firms that experienced a decrease and an increase in the marginal tax rate, respectively. Panels C and D display the effects on wage bill for firms that experienced a decrease and an increase in the marginal tax rate, respectively.

FIGURE A3. Google Searches for “*Ministerio de Hacienda*” (Ministry of Finance)



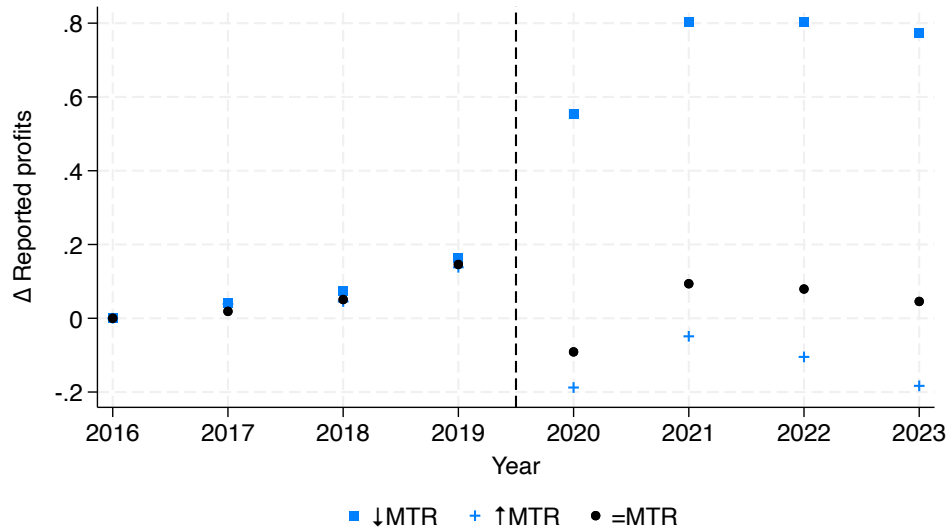
Notes: This figure displays the percentage interest for the term “Ministerio de Hacienda” (Ministry of Finance) based on Google search data from 2016 to 2022, with 100% representing peak interest. Notable spikes in search interest coincide with the announcement of the tax reform in December 2018 and its implementation in October 2019.

FIGURE A4. Difference-in-Differences by MTR Treatment Groups,
Change in Treatment Year Placebo Tests



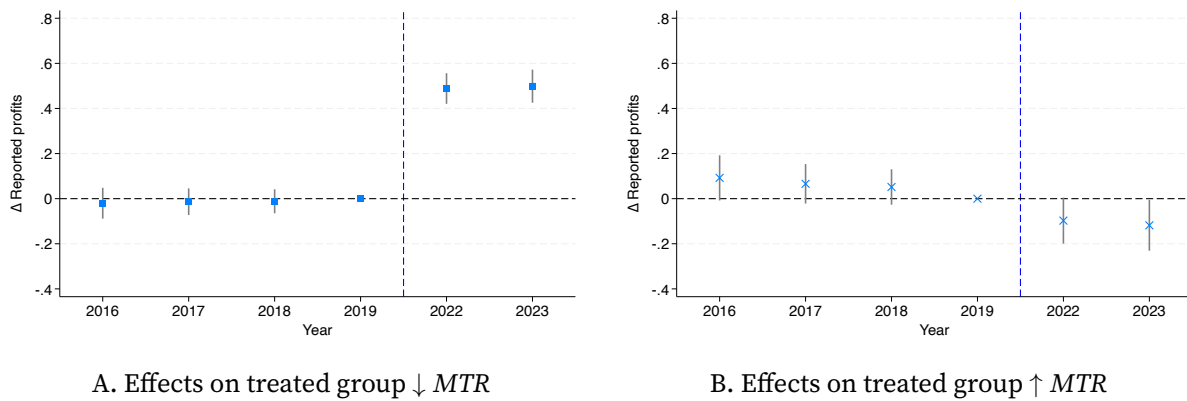
Notes: This figure presents placebo event study estimations to test the no-anticipation assumption in our difference-in-differences strategy. We replicate the main event study specifications but set 2018 and 2019 as hypothetical treatment years instead of 2020. Results show no significant effects before the actual reform year, suggesting that firms did not adjust their behaviour in anticipation of the reform. Treatment groups correspond to firms that experienced a decrease or an increase in their mechanical marginal tax rate.

FIGURE A5. Event Studies, by MTR Treatment and Control Groups



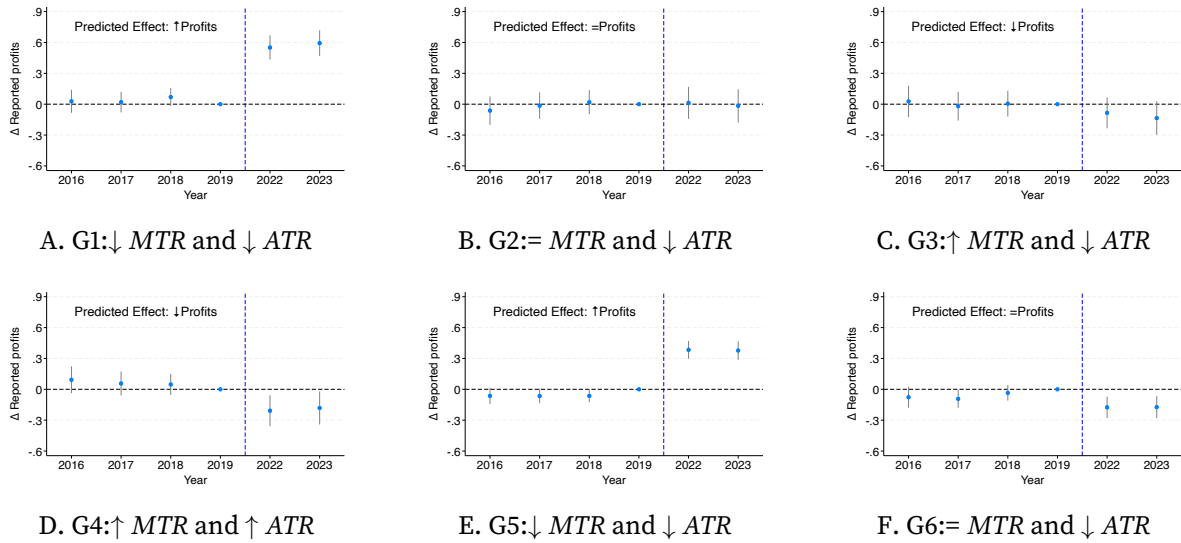
Notes: This figure presents the 2016-2023 event study estimates of reported profits for firms that experienced a decrease in the mechanical marginal tax rate, firms that experienced an increase, and firms that saw no change in the mechanical marginal tax rate. Treatment is determined based on the mechanical marginal tax rate change, calculated from pre-reform revenue and reported profit information. A Poisson quasi-maximum likelihood model is used to account for zeros in the dependent variable. The estimations include individual and time fixed effects, with standard errors clustered at the firm level.

FIGURE A6. Difference-in-Differences by MTR Treatment Groups, Robustness Excluding COVID-19 Years



Notes: This figure presents robustness checks for the event study estimations, excluding the 2020 and 2021 fiscal years, which were most affected by the COVID-19 pandemic. The treatment groups correspond to firms that experienced a decrease or an increase in their mechanical marginal tax rate. The results remain qualitatively consistent with our preferred specifications, reinforcing that the main findings are not driven by pandemic-related shocks.

FIGURE A7. Difference-in-Differences by MTR and ATR Treatment Groups,
Robustness Excluding COVID-19 Years



Notes: This figure shows the dynamics of reported profits for the six treatment groups defined by the combinations of changes in marginal and average tax rates, compared to the control group of large firms that did not experience any tax rate change. The event study estimations exclude 2020 and 2021 to address concerns about the confounding effects of COVID-19. The results confirm that the profit responses identified in the paper are not significantly affected by the pandemic period.

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